

MTH102

Linear Algebra

2025–2026–I SEMESTER

22 September 2025

Preface

This discusses the content covered in the class, along with some examples, exercises, and problem sets. You are encouraged to go through it. Solving the exercises would greatly benefit in understanding the content of the course. The exercises, remarks, marked with an asterisk (*), are supplementary. The typos/inaccuracies may be reported to jpsaha@iiserb.ac.in.

Timetable

The classes are scheduled on

- Mondays,
- Wednesdays,
- Fridays

from **11:00am to 11:55am** in L5, LHC. The tutorial is scheduled on

- Thursdays

from **5:30pm to 6:30pm** in L5, LHC.

Office hours

The office hours will be held in **Room no. 108, AB1** on

- Mondays, during **5:00pm to 6:00pm**,
- Wednesdays, during **5:00pm to 6:00pm**,
- Fridays, during **5:00pm to 6:00pm**,
- Saturdays, during **10:00am to 11:00am**.

The grading policy is as follows.

Components	Weightage
Quiz	20%
Mid-semester examination	35%
End-semester examination	45%

Calendar

- **Quiz 1** is scheduled on **Saturday, 11th October, 2025** from **3:00pm to 4:00pm**, and will be based on the content covered until **Saturday, 4th October, 2025**.
- **Quiz 2** is scheduled on **Friday, 21st November, 2025** from **11:00am to 11:55am**, and will be based on the content covered until **Saturday, 15th November, 2025**.
- The academic calendar is available at <https://www.iiserb.ac.in/doaa/schedule>.

The announcements related to the course will be made on the following Google classroom page.

<https://classroom.google.com/u/1/c/ODAwOTg3NDg2Mjg0>

The notes can be accessed at the link below (until all the students join the Google classroom).

<https://jpsaha.github.io/home/blog/2025/mth102/>

Table of contents

Preface	2
Introduction	4
1 Sets	4
1.1 Basic terminologies	6
1.2 Operations on sets	7
1.3 Family of sets	17
1.4 Power set	17
1.5 Cartesian product of sets	18
1.6 Sets and subsets of real and complex numbers	19
1.6.1 Real numbers	19
1.6.2 Complex numbers	21
2 Induction principles	24
Linear algebra	38
3 Matrices	38
3.1 Vectors	38
3.1.1 Definitions and examples	38
3.1.2 Vector operations	39
3.2 Matrix	40
3.2.1 Definitions and examples	40
3.2.2 Matrix times a vector	41
3.2.3 Matrix times a matrix	42
3.2.4 Few to many more!	49
3.3 System of linear equations	54
3.4 Invertible matrices	56
3.5 Systems of linear equations again	60
3.5.1 Linear combination of the columns of a matrix	60
3.5.2 Homogeneous system of linear equations	61
3.5.3 General system of linear equations	66
3.6 Special types of matrices	68

Introduction

Chapter 1. Sets



Definition

A **set** is a well-defined collection of objects. The objects of a set are called its **elements** or **members**.



Examples

- The set of vowels a, e, i, o, u.
- The set consisting of 2, 4, 6, 8, 10.
- The set consisting of 3, 15, 35, 63, 99.

Here are further examples.



Examples

- The set of positive integers divisible by 3.
- The set of prime numbers less than 100.
- The set of positive integers which can be expressed as the product of two distinct primes.
- The set of positive integers which can be expressed as the product of two or more distinct primes.
- The set of positive integers which are smaller than 100 and share no common factor with 100.



Remark

Note that the first few sets are described by writing down all its elements, whereas in the latter examples, the sets are described by the rules or properties which determine whether a particular object is an element or not.



Notation

A set is usually denoted by an uppercase letter, for example,

$$A, B, C, X, Y, Z,$$

whereas the lowercase letters are used to denote its elements. For instance, one may use

- a to denote an element of A ,
- b to denote an element of B ,
- c to denote an element of C ,
- x to denote an element of X ,
- y to denote an element of Y ,
- z to denote an element of Z .

Several special sets are denoted by certain standard symbols. Here are some such examples.

$\mathbb{N}$	the set of positive integers
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$\mathbb{Z}$	the set of integers
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If x is an element of a set X , then one says that x **belongs to** X , and writes

$$x \in X.$$

Note that 2 belongs to $\mathbb{N}$, but $\frac{1}{2}$ does not belong to $\mathbb{N}$. One writes

$$2 \in \mathbb{N}, \frac{1}{2} \notin \mathbb{N}.$$

Notation

There are two ways to write down a set. For example, consider the set of vowels, which is written as

$$V = \{a, e, i, o, u\}.$$

Note that the elements are separated by commas and enclosed in braces $\{\}$.

Another way to write down a set is by stating the *rules* or *properties* which determine whether a particular object is an element or not. For example, the set of even positive integers is written as

$$E = \{x \in \mathbb{Z} : x \text{ is even}, x > 0\}.$$

This reads

“ E is the set of elements x in $\mathbb{Z}$ such that x is even and $x > 0$ ”.

Example

Note that the set $A = \{3, 15, 35, 63, 99\}$ is equal to

$$\{x \in \mathbb{Z} : x \text{ is equal to } n^2 - 1 \text{ for some } n \in \{2, 4, 6, 8, 10\}\}.$$

Since 1, 2 are the roots of the polynomial $x^2 - 3x + 2$, it follows that

$$\{x \in \mathbb{N} : x^2 - 3x + 2 = 0\} = \{1, 2\}.$$

Also note that

$$\mathbb{N} = \{1, 2, 3, \dots\},$$

$$\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \dots\}.$$

One uses the symbol $:=$ to indicate that the symbol on the left is being defined by the symbol on the right.

$$\mathbb{Q} := \left\{ \frac{m}{n} : m, n \in \mathbb{Z} \text{ and } n \neq 0 \right\}.$$

It is called the *set of rational numbers*.

If the number of elements of a set is finite, then it is called a **finite set**. If a set is not finite, it is called an **infinite set**. A set containing exactly one element is called a **singleton set**.

Example

The sets

$$\{a, e, i, o, u\}, \{3, 15, 35, 63, 99\}, \{x \in \mathbb{N} : x^2 - 3x + 2 = 0\}$$

are finite. The sets $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are infinite.

Exercise 1.1. Determine the elements of the following sets.

- $\{x \in \mathbb{N} : x^2 - 1 = 0\}.$
- $\{x \in \mathbb{Z} : x^2 - 1 = 0\}.$

Solution. Note that 1 is the only element of $\mathbb{N}$ satisfying the equation $x^2 - 1 = 0$. This gives

$$\{x \in \mathbb{N} : x^2 - 1 = 0\} = \{1\}.$$

Since 1, -1 are precisely all the elements of $\mathbb{Z}$ satisfying $x^2 - 1 = 0$, it follows that

$$\{x \in \mathbb{Z} : x^2 - 1 = 0\} = \{1, -1\}.$$

□

§1.1 Basic terminologies

If A, B are sets, and every element of A also belongs to B , then we say that A is a **subset** of B , and write

$$A \subseteq B.$$

If A is not a subset of B , one writes

$$A \not\subseteq B.$$



Example

- Note that every set is a subset of itself.
- Note that $\mathbb{N}$ is a subset of $\mathbb{Z}$, and $\mathbb{Z}$ is a subset of $\mathbb{Q}$. One writes

$$\mathbb{N} \subseteq \mathbb{Z}, \quad \mathbb{Z} \subseteq \mathbb{Q}.$$

- Also note that

$$\mathbb{Z} \not\subseteq \mathbb{N}, \mathbb{Q} \not\subseteq \mathbb{Z}, \mathbb{Q} \not\subseteq \mathbb{N}.$$

If A is a subset of B and A is not equal to B , then A is said to be a *proper subset* of B , and one writes

$$A \subsetneq B.$$

Note that

$$\mathbb{N} \subsetneq \mathbb{Z}, \mathbb{Z} \subsetneq \mathbb{Q}.$$

Exercise 1.2. What does it mean to say that A is not a subset of B ?

Solution. Note that A is a subset of B is equivalent to the statement that each element of A lies in B . Hence, A is not a subset of B is equivalent to saying that some element of A does not belong to B . □

Exercise 1.3. Show that if A is a subset of B and B is a subset of C , then A is a subset of C .

Solution. Let x be an element of A . Since A is a subset of B , it follows that x belongs to B . Using that B is a subset of C , we obtain that x belongs to C . Hence, the element x lies in C . This shows that any element x of A belongs to C . This proves that A is a subset of C . □

Exercise 1.4. Show that no element n of $\mathbb{N}$ satisfies $n^4 - 5n^2 + 6 = 0$.



Tip

Try to show that no natural number n satisfies any of the equations

$$n^2 = 2, n^2 = 3.$$

Solution. Note that the polynomial $x^4 - 5x^2 + 6$ factorizes as

$$x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3).$$

Let n be a natural number. If $n = 1$, then $n^2 \neq 2$ and $n^2 \neq 3$ hold. If $n \geq 2$, then $n^2 \geq 4$, and hence, $n^2 \neq 2$ and $n^2 \neq 3$ hold. This shows that no natural number n satisfies

$$(n^2 - 2)(n^2 - 3) = 0,$$

that is, the equation

$$n^4 - 5n^2 + 6 = 0.$$

□

Consider the set

$$A = \{n \in \mathbb{N} : n^4 - 5n^2 + 6 = 0\}.$$

Note that the set A has no elements. Any such set is called the **empty set** or **null set**, and is denoted by the symbol

$$\emptyset.$$

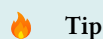
Two sets, A and B , are said to be **equal**¹ if any element of A belongs to B , and any element of B also belongs to A , or equivalently,

$$A \subseteq B \text{ and } B \subseteq A$$

holds. If two sets A, B are not equal, one writes

$$A \neq B.$$

Exercise 1.5. What does it mean to say that two sets A, B are not equal?



Tip

Does [Exercise 1.2](#) help?

Solution. Note that two sets A, B are equal is equivalent to saying that A is a subset of B , and B is a subset of C . Hence, the statement $A \neq B$ is equivalent to saying that A is **not** a subset of B , **or** B is **not** a subset of A , which is equivalent to the statement that some element of A does not belong to B , **or** some element of B does not belong to A . □

§1.2 Operations on sets

Exercise 1.6. Consider the sets

$$\{1, 2, 3, 4\}, \{3, 4, 5, 6\}.$$

Determine the elements common to these sets.

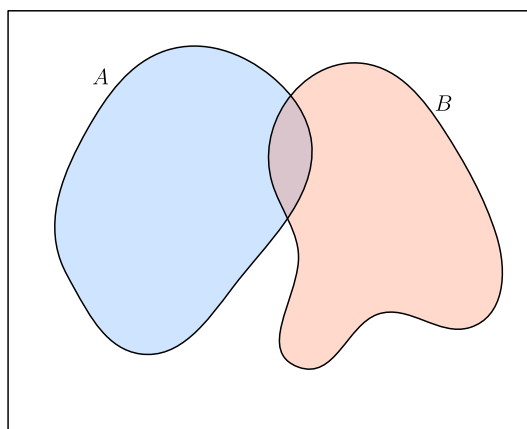


Figure 1: Two sets A and B

¹It may appear obvious! The advantage of putting forth definitions is to set up notations and conventions, so that these are not left to interpretations!

 Definition

If A, B are sets, their **union** is denoted by $A \cup B$, which is defined as

$$A \cup B := \{x : x \in A \text{ or } x \in B\}.$$

See [Figure 2](#).

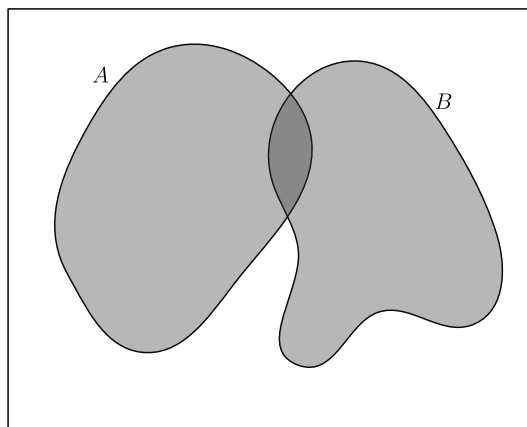


Figure 2: The union of A and B

Exercise 1.7. Determine the union of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

 Usage of “or” in the inclusive sense

The word “or” is used in the inclusive sense, allowing two or more of the conditions to be satisfied. In common terminology, this inclusive sense is denoted by “and/or”.

Exercise 1.8. If A, B are sets, show that

$$A \subseteq A \cup B, B \subseteq A \cup B.$$

Solution. Let x be an element of A . Then note that x lies $A \cup B$. This shows that A is a subset of $A \cup B$. Similarly, it follows that B is a subset of $A \cup B$. $\square$

Exercise 1.9. If A, B are subsets of a set C , then $A \cup B$ is a subset of C .

Solution. Let x be an element of $A \cup B$. It follows that x belongs to A or x belongs to B . If x belongs to A , then using A is a subset of C , we obtain that x lies in C . Further, if x belongs to B , then using that B is a subset of C , we obtain that x lies in C . Combining the above cases, it follows that x lies in C . Consequently, any element of $A \cup B$ is an element of C , or equivalently, $A \cup B$ is a subset of C . $\square$

Exercise 1.10. If A is a set, show that

$$A \cup A = A.$$

Solution. Note that any element x of $A \cup A$ belongs to A or belongs to A , and hence it lies in A . This shows that $A \cup A$ is a subset of A . Further, for any element y of A , it belongs to $A \cup A$. Hence, A is a subset of $A \cup A$. This proves that

$$A \cup A = A.$$

$\square$

Exercise 1.11 (Commutative property). If A, B are sets, show that

$$A \cup B = B \cup A.$$

Solution. Let P, Q be sets, and let x be an element of $P \cup Q$. It follows that x belongs to P or x belongs to Q . This shows that x belongs to Q or x belongs to P . This implies that x is an element of $Q \cup P$. Hence, $P \cup Q$ is a subset of $Q \cup P$ for any two sets P, Q .

Consequently, the set $A \cup B$ is a subset of $B \cup A$, and $B \cup A$ is a subset of $A \cup B$. This shows that

$$A \cup B = B \cup A.$$

□



Definition

If A, B, C are sets, their **union** is denoted by $A \cup B \cup C$, which is defined as

$$A \cup B \cup C := \{x : x \in A \text{ or } x \in B \text{ or } x \in C\}.$$

Exercise 1.12 (Associative property). If A, B, C are sets, show that

$$A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C).$$

Use [Exercise 1.11](#) and [Exercise 1.12](#), to show that for any three sets A, B, C , the following sets are equal $A \cup B \cup C, A \cup C \cup B, B \cup A \cup C, B \cup C \cup A, C \cup A \cup B, C \cup B \cup A$.



Definition

If A, B are sets, their **intersection** is denoted by $A \cap B$, which is defined as

$$A \cap B := \{x : x \in A \text{ and } x \in B\}.$$

See [Figure 3](#).

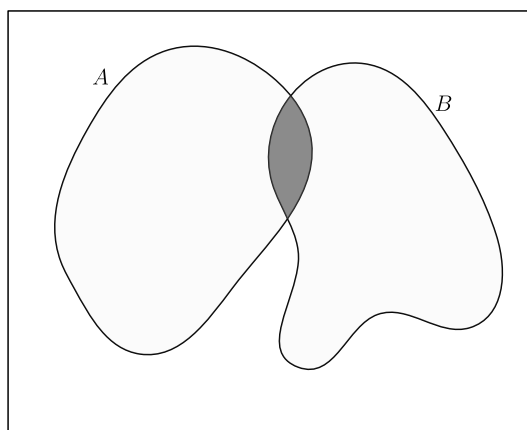


Figure 3: The intersection of A and B

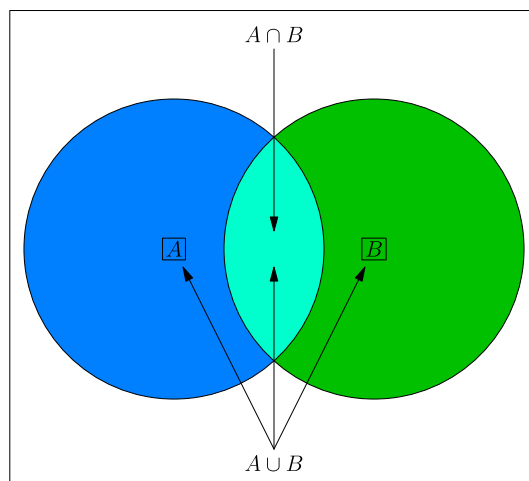


Figure 4: The union and intersection of A and B

Exercise 1.13. Determine the intersection of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Exercise 1.14. If A, B are sets, show that

$$A \cap B \subseteq A, A \cap B \subseteq B.$$

Solution. For any element x of $A \cap B$, it belongs to A and it belongs to B . Hence, $A \cap B$ is a subset of A , and $A \cap B$ is also a subset of B . $\square$

Exercise 1.15. If A, B, C are sets satisfying

$$C \subseteq A, C \subseteq B,$$

then show that

$$C \subseteq A \cap B$$

holds.

Exercise 1.16. Identify the integers among

$$1, 2, 3, 4, \dots, 20$$

which are divisible by at least one of 2 and 5.

Solution. In the following, the integers among $1, 2, \dots, 20$ divisible by 2 are marked.

1, $\textcircled{2}$, 3, $\textcircled{4}$, 5, $\textcircled{6}$, 7, $\textcircled{8}$, 9, $\textcircled{10}$,
11, $\textcircled{12}$, 13, $\textcircled{14}$, 15, $\textcircled{16}$, 17, $\textcircled{18}$, 19, $\textcircled{20}$.

Note that there 10 integers among $1, 2, \dots, 20$, which are divisible by 2. The integers among $1, 2, \dots, 20$ divisible by 5 are marked below.

1, 2, 3, 4, $\textcircled{5}$, 6, 7, 8, 9, $\textcircled{10}$,
11, 12, 13, 14, $\textcircled{15}$, 16, 17, 18, 19, $\textcircled{20}$.

Note that there are 4 integers among $1, 2, \dots, 20$, which are divisible by 5. $\square$

 Remark

Note that some integers are counted/marked twice. This is an instance of *double counting*, which refers to the counting of certain element twice. In the above, the integers 10 and 20 are “counted twice”. Thus, the number of integers among $1, 2, \dots, 20$, divisible by 2 or 5, is equal to $10 + 4 - 2 = 12$.

 Definition

Two sets A, B are said to be **disjoint** if

$$A \cap B = \emptyset.$$

Lemma 1.17 (Inclusion-exclusion principle). *Let X be a set, and A, B be finite subsets of X . Then*

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Proof. Note that the set $A \cup B$ is equal to the union of the disjoint sets A and $B \setminus A$. This shows that

$$|A \cup B| = |A| + |B \setminus A|.$$

Also note that the set B is equal to the union of the disjoint subsets $A \cap B$ and $B \setminus A$. This gives

$$|B| = |A \cap B| + |B \setminus A|.$$

It follows that

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

□

Exercise 1.18 (Inclusion-exclusion principle). Using [Lemma 1.17](#) or otherwise, show that for finite subsets A, B, C of a set X ,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

Solution. Note that

$$\begin{aligned} & |A \cup B \cup C| \\ &= |(A \cup B) \cup C| \\ &= |A \cup B| + |C| - |(A \cup B) \cap C| && \text{(by Lemma 1.17)} \\ &= |A \cup B| + |C| - |(A \cap C) \cup (B \cap C)| && \text{(by Exercise 1.23)} \\ &= |A \cup B| + |C| - (|A \cap C| + |B \cap C| - |(A \cap C) \cap (B \cap C)|) && \text{(by Lemma 1.17)} \\ &= |A \cup B| + |C| - |A \cap C| - |B \cap C| + |(A \cap C) \cap (B \cap C)| \\ &= |A \cup B| + |C| - |A \cap C| - |B \cap C| + |A \cap C \cap B \cap C| \\ &= |A \cup B| + |C| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \\ &= |A| + |B| - |A \cap B| + |C| - |A \cap C| - |B \cap C| + |A \cap C \cap B \cap C| && \text{(by Lemma 1.17)} \\ &= |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|. \end{aligned}$$

□

Exercise 1.19. Determine the number of integers among

$$1, 2, 3, 4, \dots, 100,$$

which are divisible by at least one of 6, 10, 15.

Solution. Let X denote the set

$$\{1, 2, 3, \dots, 100\}.$$

Let A (resp. B, C) denote the set of elements X which are divisible by 6 (resp. 10, 15). Note that

$$|A| = 16, |B| = 10, |C| = 6$$

hold. Observe that $A \cap B$ consists of the elements of X which are divisible by 6 and by 10, that is, divisible by their least common multiple, which is equal to 30. Similarly, $B \cap C$ consists of the elements of X which are divisible by the least common multiple of 10, 15, which is equal to 30. Further, it follows that $C \cap A$ also consists of the multiples of 30 lying between 1 and 100. Moreover, the set $A \cap B \cap C$ consists of the integers between 1 and 100, which are divisible by the least common multiple of 6, 10, 15, that is, the integer 30. By the inclusion-exclusion principle ([Exercise 1.18](#)), we obtain

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C| \\ &= |A| + |B| + |C| - 3|A \cap B| + |A \cap B \cap C| \\ &= |A| + |B| + |C| - 2|A \cap B| \\ &= 16 + 10 + 6 - 2 \times 3 \\ &= 26. \end{aligned}$$

Note that the set $A \cup B \cup C$ consists of the integers between 1 and 100 which are divisible by at least one of 6, 10, 15. This shows that there are precisely **26** such integers. $\square$

Exercise 1.20. If A is a set, show that

$$A \cap A = A.$$



Tip

Does [Exercise 1.10](#) help? If not, what about its solution?

Exercise 1.21 (Commutative property). If A, B are sets, show that

$$A \cap B = B \cap A.$$



Tip

Does [Exercise 1.11](#) help? If not, what about its solution?



Definition

If A, B, C are sets, their **intersection** is denoted by $A \cap B \cap C$, which is defined as

$$A \cap B \cap C := \{x : x \in A \text{ and } x \in B \text{ and } x \in C\}.$$

Exercise 1.22 (Associative property). If A, B, C are sets, show that

$$A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C).$$



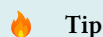
Tip

Does [Exercise 1.12](#) help? If not, what about its solution?

Exercise 1.23 (Distributive property). If A, B, C are sets, show that

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

**Tip**

Recall when two sets are equal.

Solution. Note that if P is a subset of Q , then we claim that $A \cup P$ is a subset of $A \cup Q$. Indeed, for an element x of $A \cup P$, note that x lies in A , or it lies in P . If x lies in A , then it lies in $A \cup Q$. If x lies in P , it also lies in Q , and hence it lies in $A \cup Q$. This proves the claim. Consequently, $A \cup (B \cap C)$ is a subset of both of the sets

$$A \cup B, A \cup C,$$

and hence, we obtain

$$A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C).$$

Let y be an element of $(A \cup B) \cap (A \cup C)$. Note that y lies in $A \cup B$ and y also lies in $A \cup C$. If y lies in A , then it also lies in $A \cup (B \cap C)$. Thus, it remains to consider the case that y does not lie in A , which we assume from now on. Since y lies in $A \cup B$ and y does not lie in A , it follows that y lies in B . Similarly, it also follows that y lies in C . This shows that y lies in $B \cap C$, and hence it lies in $A \cup (B \cap C)$. This proves that

$$(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C).$$

This establishes that

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Let us now establish that

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Let x be an element of $A \cap (B \cup C)$. Note that x lies in A and also lies in $B \cup C$. If x lies in B , then x lies in $A \cap B$, and hence x belongs to $(A \cap B) \cup (A \cap C)$. It remains to consider the case that x does not lie in B , which we assume from now on. Since x lies in $B \cup C$ and x does not lie in B , it follows that x lies in C . Using that x lies in A , we obtain that x lies in $A \cap C$, and hence, x lies in $(A \cap B) \cup (A \cap C)$. Combining these cases, we obtain

$$A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C).$$

Note that if P is a subset of Q , then $A \cap P$ is contained in $A \cap Q$. It follows that $A \cap B$ is a subset of $A \cap (B \cup C)$, and $A \cap C$ is also a subset of $A \cap (B \cup C)$. This shows that

$$(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C).$$

This proves that

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

□

 Remark

All sets under consideration in a given context, are assumed to be contained in a ‘large’ fixed set, called the *universal set*. For example, while studying the positive integers (for instance, some sets consisting of the positive integers), the set $\mathbb{N}$ can be taken as the universal set. While studying the integers, the set $\mathbb{Z}$ can be taken as the universal set.

 Definition

Let A, B be subsets of a set X . The **complement of A in B** is denoted by $B \setminus A$, and is defined by

$$B \setminus A := \{x \in X : x \in B, x \notin A\}$$

It is also called the **difference of B and A** .

The complement of A in X , is called the *complement of A* , and is denoted by A^c , and is defined to be

$$A^c := X \setminus A.$$

Exercise 1.24. Show that

$$A^c = \{x \in X : x \notin A\},$$

where X denotes the underlying universal set.

Solution. Since $X \setminus A$ is equal to $\{x \in X : x \notin A\}$, it follows that

$$A^c = \{x \in X : x \notin A\}.$$

□

Exercise 1.25. Determine the complement of $\{1, 2, 3, 4\}$ in $\{1, 3, 5, 7\}$, and the complement of $\{1, 3, 5, 7\}$ in $\{1, 2, 3, 4\}$.

Exercise 1.26. If A, B are subsets of a set X , show that

$$A^c \cap B = B \setminus A.$$

Solution. Note that

$$\begin{aligned} A^c \cap B &= (X \setminus A) \cap B \\ &= \{x \in X : x \in X \setminus A \text{ and } x \in B\} \\ &= \{x \in X : x \notin A \text{ and } x \in B\} \\ &= \{x \in X : x \in B \text{ and } x \notin A\} \\ &= B \setminus A. \end{aligned}$$

□

Exercise 1.27. If A is a subset of a set X , then show that

$$A \cup A^c = X, A \cap A^c = \emptyset, (A^c)^c = A.$$

Exercise 1.28. If A, B are subsets of a set X , then show that

$$(A \cup B)^c = A^c \cap B^c, (A \cap B)^c = A^c \cup B^c.$$

Solution. Note that

$$\begin{aligned}
(A \cup B)^c &= X \setminus (A \cup B) \\
&= \{x \in X : x \notin (A \cup B)\} \\
&= \{x \in X : x \notin A \text{ and } x \notin B\} \\
&= \{x \in X : x \in A^c \text{ and } x \in B^c\} \\
&= \{x \in X : x \in A^c \cap B^c\} \\
&= A^c \cap B^c
\end{aligned}$$

holds for any subsets A, B of X .

As a consequence, we obtain

$$(A^c \cup B^c)^c = ((A^c)^c \cap (B^c)^c),$$

which yields

$$(A^c \cup B^c)^c = A \cap B.$$

This implies

$$((A^c \cup B^c)^c)^c = (A \cap B)^c,$$

or equivalently,

$$(A \cap B)^c = A^c \cup B^c.$$

□

Exercise 1.29. Let A, B be subsets of a set X . Show that the following statements are equivalent.

$$\begin{aligned}
&A \subseteq B, \\
&A \cap B = A, \\
&A \cup B = B, \\
&B^c \subseteq A^c.
\end{aligned}$$

Exercise 1.30. If A, B, C are sets, show that

1. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C),$
2. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$



Tip

Do the above look familiar in the special case when A is equal to the universal set?

Solution. Let X denote the underlying universal set. Note that

$$\begin{aligned}
X \setminus (B \cup C) &= (B \cup C)^c \\
&= B^c \cap C^c \\
&= (X \setminus B) \cap (X \setminus C).
\end{aligned}$$

This shows that

$$A \cap (X \setminus (B \cup C)) = A \cap ((X \setminus B) \cap (X \setminus C)),$$

which implies that

$$A \cap (X \setminus (B \cup C)) = (A \cap (X \setminus B)) \cap (A \cap (X \setminus C)).$$

Note that for any subset Y of X , we have

$$\begin{aligned}
A \cap (X \setminus Y) &= \{x \in X : x \in A \text{ and } x \in X \setminus Y\} \\
&= \{x \in X : x \in A \text{ and } x \notin Y\} \\
&= A \setminus Y.
\end{aligned}$$

Consequently, we obtain

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$

Similarly, we obtain

$$\begin{aligned}
X \setminus (B \cap C) &= (B \cap C)^c \\
&= B^c \cup C^c \\
&= (X \setminus B) \cup (X \setminus C).
\end{aligned}$$

This shows that

$$A \cap (X \setminus (B \cap C)) = A \cap ((X \setminus B) \cup (X \setminus C)).$$

Using the distributive property of intersection over union (see [Exercise 1.23](#)), we obtain

$$A \cap (X \setminus (B \cap C)) = (A \cap (X \setminus B)) \cup (A \cap (X \setminus C)).$$

Since for any subset Y of X , the equality

$$A \cap (X \setminus Y) = A \setminus Y$$

holds, we conclude that

$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$$

□

Exercise 1.31. Let A, B be subsets of a set X . Show that the sets

$$A \setminus B, B \setminus A$$

are disjoint.



Definition

The **symmetric difference** of two sets A, B , denoted by $A \Delta B$, is defined as

$$A \Delta B := (A \cup B) \setminus (A \cap B).$$

Exercise 1.32. Determine the symmetric difference of $\{1, 2, 3, 4\}$ and $\{1, 3, 5, 7\}$.

Exercise 1.33. If A, B are sets, then show that

$$\begin{aligned}
A \Delta B &= (A \setminus B) \cup (B \setminus A), \\
A \Delta B &= B \Delta A, \\
A \cup B &= (A \Delta B) \cup (A \cap B), \\
(A \Delta B) \cap (A \cap B) &= \emptyset.
\end{aligned}$$

Solution. Note that

$$\begin{aligned}
A \Delta B &= (A \cup B) \setminus (A \cap B) \\
&= (B \cup A) \setminus (B \cap A) \\
&= B \Delta A.
\end{aligned}$$

Since $A \cap B$ is a subset of $A \cup B$, we obtain

$$\begin{aligned} A \cup B &= ((A \cup B) \setminus (A \cap B)) \cup (A \cap B) \\ &= (A \Delta B) \cup (A \cap B). \end{aligned}$$

Using $A \Delta B = (A \cup B) \setminus (A \cap B)$, it follows that $A \Delta B$ contains no element of $A \cap B$. This implies that

$$(A \Delta B) \cap (A \cap B) = \emptyset.$$

□

Exercise 1.34 ().** Do there exist three finite sets A, B, C satisfying

$$|A \Delta B| = 1, |B \Delta C| = 2, |C \Delta A| = 4?$$

Solution. Note that for any three sets A, B, C ,

$$\begin{aligned} C \setminus A &= (C \setminus A) \cap (B \cup B^c) \\ &= ((C \setminus A) \cap B) \cup ((C \setminus A) \cap B^c) \\ &\subseteq (A^c \cap B) \cup (C \cap B^c) \\ &= (B \setminus A) \cup (C \setminus B) \\ &\subseteq (A \Delta B) \cup (B \Delta C) \end{aligned}$$

hold. Hence, for any three sets A, B, C , we also obtain

$$\begin{aligned} A \setminus C &\subseteq (C \Delta B) \cup (B \Delta A) \\ &= (A \Delta B) \cup (B \Delta C). \end{aligned}$$

This shows that for any three sets A, B, C , the union of $A \setminus C, C \setminus A$ is a subset of $(A \Delta B) \cup (B \Delta C)$, that is, $C \Delta A$ is a subset of $(A \Delta B) \cup (B \Delta C)$. If A, B, C are finite sets, it follows that

$$|C \Delta A| \leq |(A \Delta B) \cup (B \Delta C)| \leq |A \Delta B| + |B \Delta C|.$$

This shows that there are no three finite sets A, B, C satisfying the given conditions. □

§1.3 Family of sets



Definition

Let $n \geq 2$ be an integer, and let $A_1, \dots, A_n$ be sets. The **union of $A_1, \dots, A_n$** is denoted by $A_1 \cup \dots \cup A_n$, and is defined by

$$A_1 \cup \dots \cup A_n := \{x : x \text{ belongs to } A_i \text{ for some } 1 \leq i \leq n\}.$$

The **intersection of $A_1, \dots, A_n$** is denoted by $A_1 \cap \dots \cap A_n$, and is defined by

$$A_1 \cap \dots \cap A_n := \{x : x \text{ belongs to } A_i \text{ for all } 1 \leq i \leq n\}.$$

§1.4 Power set



Example

Note that the subsets of $\{1, 2, 3\}$ are

$$\begin{aligned} &\{1, 2, 3\}, \\ &\{1, 2\}, \{2, 3\}, \{3, 1\}, \\ &\{1\}, \{2\}, \{3\}, \\ &\emptyset. \end{aligned}$$

**Definition**

Let A be a set. The set of subsets of A is denoted by $\mathcal{P}(A)$. It is called the **power set of A** .

**Example**

Note that

$$\mathcal{P}(\{1\}) = \{\emptyset, \{1\}\},$$

$$\mathcal{P}(\{x\}) = \{\emptyset, \{x\}\},$$

$$\mathcal{P}(\{x, y\}) = \{\emptyset, \{x\}, \{y\}, \{x, y\}\},$$

$$\mathcal{P}(\{x, y, z\}) = \{\emptyset, \{x\}, \{y\}, \{z\}, \{x, y\}, \{y, z\}, \{z, x\}, \{x, y, z\}\},$$

$$\mathcal{P}(\{1, 2, 3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}\},$$

$$\mathcal{P}(\{1, 2, 3, 4\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}, \\ \{4\}, \{1, 4\}, \{2, 4\}, \{3, 4\}, \{1, 2, 4\}, \{2, 3, 4\}, \{1, 3, 4\}, \{1, 2, 3, 4\}\}.$$

**Remark**

Convince yourself that for any integer $n \geq 0$, the power set of a set of size n has size 2^n .

§1.5 Cartesian product of sets

**Definition**

If A, B are sets, then the **cartesian product** $A \times B$ of A and B is the set of all ordered pairs of the form (a, b) with $a \in A$ and $b \in B$. That is,

$$A \times B := \{(a, b) : a \in A, b \in B\}.$$

If $A = \{1, 2\}$ and $B = \{2, 3, 4\}$, then

$$A \times B = \{(1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4)\}.$$

Exercise 1.35. If A, B are finite sets, then show that

$$|A \times B| = |A| \times |B|,$$

where for a set X , its number of elements is denoted by $|X|$.

§1.6 Sets and subsets of real and complex numbers

§1.6.1 Real numbers

$\mathbb{N}$ = the set of positive integers

$$= \{1, 2, 3, \dots\},$$

$\mathbb{Z}$ = the set of integers

$$= \{0, 1, -1, 2, -2, 3, -3, \dots\},$$

$\mathbb{Q}$ = the set of rational numbers

$$= \left\{ \frac{m}{n} : m \in \mathbb{Z}, n \in \mathbb{N} \right\},$$

$\mathbb{R}$ = the set of real numbers.

i Remark

There are real numbers which are not rational. For instance, the number $\sqrt{2}$ is not rational.

Lemma 1.36. *No rational number x satisfies $x^2 = 2$.*

Proof. On the contrary, let us assume that some rational number x satisfies $x^2 = 2$. Replacing x by $-x$ if necessary, we may and do assume that x is positive. Write $x = \frac{a}{b}$ for some positive integers a, b . Let d denote the greatest common divisor of a, b . Write $a = da_0, b = db_0$ where a_0, b_0 are positive integers. Note that $\left(\frac{a_0}{b_0}\right)^2 = 2$.

Let us prove the two claims below. After establishing them, we will use them to complete the proof of the lemma.

Claim

None of a_0, b_0 is divisible by 3.

Proof of the Claim

On the contrary, let us assume that 3 divides at least one of the integers a_0, b_0 .

If 3 divides a_0 , then 3 divides a_0^2 . Using $a_0^2 = 2b_0^2$, it follows that 3 divides b_0 . Further, if 3 divides b_0 , then 3 divides $2b_0^2$, and hence 3 divides a_0^2 . This shows that a_0 is divisible by 3.

In both the cases, it follows that the greatest common divisor of a_0, b_0 is larger than 1. This shows that the greatest common divisor of da_0, db_0 is larger than d . Using $a = da_0, b = db_0$, we obtain that the greatest common divisor of a, b is larger than d , which is impossible. This contradicts the assumption that some rational number x satisfies $x^2 = 2$. This proves the claim.

Claim

If n is an integer not divisible by 3, then n^2 is equal to $3k + 1$ for some integer k , depending on n .

Proof of the Claim

Let q (resp. r) denote the quotient (resp. remainder) obtained upon dividing n by 3. This gives $n = 3q + r$. Note that

$$\begin{aligned} n^2 &= (3q + r)^2 \\ &= 9q^2 + 6qr + r^2 \end{aligned}$$

Since $n = 3q + r$ and 3 does not divide n , it follows that r is equal to 1 or 2. If $r = 1$, then

$$n^2 = 3(3q^2 + 2qr) + 1.$$

If $r = 2$, then

$$n^2 = 3(3q^2 + 2qr) + 4 = 3(3q^2 + 2qr + 1) + 1.$$

This proves the claim.

Using the claims above, it follows that

$$\begin{aligned} a_0^2 &= 3c + 1, \\ b_0^2 &= 3d + 1 \end{aligned}$$

for some integers c, d . This gives

$$3c + 1 = 2(3d + 1).$$

This yields $1 = 3(c - 2d)$, which is impossible.

This completes the proof of the lemma. □

**Definition**

Let a, b be real numbers with $a \leq b$. Then the **closed interval** $[a, b]$ is defined by

$$[a, b] := \{x \in \mathbb{R} : a \leq x \leq b\}.$$

The **open interval** (a, b) is defined by

$$(a, b) := \{x \in \mathbb{R} : a < x < b\}.$$

Exercise 1.37. Determine the cartesian product of $[1, 2]$ and $[3, 4] \cup [5, 6]$.

Exercise 1.38. Identify the set

$$\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\}.$$

Solution. Let x be an element of $\mathbb{R} \setminus \{0\}$. Note that

$$\begin{aligned} x + \frac{1}{x} - 2 &= \frac{x^2 + 1 - 2x}{x} \\ &= \frac{(x - 1)^2}{x} \end{aligned}$$

hold. This shows that the condition

$$x + \frac{1}{x} \geq 2$$

is equivalent to the condition

$$\frac{(x - 1)^2}{x} \geq 0. \tag{1}$$

If $x \neq 1$, then we obtain $(x - 1)^2 > 0$, and then, Equation 1 yields $\frac{1}{x} > 0$, or equivalently, $x > 0$. This gives that

$$\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\} \setminus \{1\} \subseteq \{x \in \mathbb{R} : x > 0\},$$

which implies

$$\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\} \subseteq \{x \in \mathbb{R} : x > 0\}.$$

Also note that if $x > 0$, then

$$\frac{(x - 1)^2}{x} \geq 0,$$

or equivalently,

$$x + \frac{1}{x} \geq 2.$$

It follows that

$$\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\} = \{x \in \mathbb{R} : x > 0\}.$$

□

Exercise 1.39. Let A, B be subsets of $\mathbb{R}$ defined by

$$A = \{x \in \mathbb{R} : x^2 \geq 0\},$$

$$B = \{x \in \mathbb{R} : x^3 \geq 0\}.$$

Determine the sets $A \cup B$, $A \cap B$, $A \setminus B$, $B \setminus A$.

Exercise 1.40. For a positive integer k , let A_k denote the set of integral multiples of k , that is,

$$A_k := \{r \in \mathbb{Z} : r = k\ell \text{ for some } \ell \in \mathbb{Z}\}.$$

Let m, n be positive integers. For each of the following statements, determine the equivalent conditions on the integers m, n .

1. $A_m \subseteq A_n$
2. $A_m \subsetneq A_n$
3. $A_m \not\subseteq A_n$
4. $A_m = A_n$
5. $A_m \cap A_n = \emptyset$
6. $A_m \setminus A_n \neq \emptyset$
7. $A_m \setminus A_n \neq \emptyset$ or $A_n \setminus A_m \neq \emptyset$
8. $A_m \setminus A_n \neq \emptyset$ and $A_n \setminus A_m \neq \emptyset$

§1.6.2 Complex numbers

Note that no real number x satisfies $x^2 = -1$. To “resolve” this, one “introduces”² an element i satisfying

$$i^2 = -1.$$

²To introduce is to adjoin.

The element i can be multiplied by any real number, and the product can also be added to any real number. This leads to the numbers of the form $a + ib$, where a, b are real numbers. Such numbers are called the **complex numbers**, and the set of such numbers is denoted by $\mathbb{C}$.

Definition

The **set of complex numbers**, is denoted by $\mathbb{C}$, and is defined by

$$\mathbb{C} := \{a + ib : a, b \in \mathbb{R}\}.$$

Exercise 1.41. Show that $\mathbb{R}$ is a subset of $\mathbb{C}$.

Definition

For a complex number $z = a + ib$ with $a, b \in \mathbb{R}$, the **real part** (resp. **imaginary part**) of z , denoted by $\Re(z)$ (resp. $\Im(z)$), is defined by

$$\Re(z) = a,$$

$$\Im(z) = b.$$

Definition

For complex numbers $z = a + ib, w = c + id$ with $a, b, c, d \in \mathbb{R}$, the **sum** $z + w$ and the **product** $z \cdot w$ of z and w are defined by

$$z + w := (a + c) + i(b + d),$$

$$z \cdot w := (ac - bd) + i(ad + bc).$$

Definition

For a complex number $z = a + ib$ with $a, b \in \mathbb{R}$, the **conjugate** of z , denoted by $\bar{z}$, is defined by

$$\bar{z} := a - ib.$$

Definition

For a complex number $z = a + ib$ with $a, b \in \mathbb{R}$, the **modulus** of z , denoted by $|z|$, is defined by

$$|z| := \sqrt{a^2 + b^2}.$$

Exercise 1.42. If z, w are complex numbers, then show that

1. $\overline{z + w} = \bar{z} + \bar{w}$,
2. $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$,
3. $|z| \geq 0$, and $|z| = 0$ if and only if $z = 0$,
4. $|z \cdot w| = |z| |w|$,
5. $|z + w| \leq |z| + |w|$ (triangle inequality).

Exercise 1.43. If z is a complex number, then show that

$$z \cdot \bar{z} = |z|^2.$$

i Remark ()**

More formally, one defines $\mathbb{C}$, as the following set

$$\{(a, b) : a, b \in \mathbb{R}\},$$

equipped with addition and multiplication defined by

$$(a, b) + (c, d) := (a + c, b + d),$$

$$(a, b) \cdot (c, d) := (ac - bd, ad + bc),$$

for any $(a, b), (c, d)$ with $a, b, c, d \in \mathbb{R}$.

Exercise 1.44. For any real number θ , show that

$$|\cos \theta + i \sin \theta| = 1.$$

Exercise 1.45. If z is a complex number satisfying $|z| = 1$, then show that there exists a real number θ such that

$$z = \cos \theta + i \sin \theta.$$

Exercise 1.46. If z is a nonzero complex number, then show that

$$z = |z| (\cos \theta + i \sin \theta)$$

for some real number θ .

Chapter 2. Induction principles



Example

Show that

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

holds for any positive integer n .



Solution

For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Note that $P(1)$ holds. Let k be an element of $\mathbb{N}$, and assume that $P(k)$ holds. Note that

$$\begin{aligned} 1 + 2 + 3 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \quad (\text{using } P(k)) \\ &= (k+1) \left(\frac{k}{2} + 1 \right) \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n .



Definition

If A is a nonempty subset of $\mathbb{N}$, then an element a_0 of A is called a **least element** of A if $a_0 \leq a$ for all $a \in A$.

Exercise 2.1. If A is a nonempty subset of $\mathbb{N}$, and a_1, a_2 are least elements of A , then show that $a_1 = a_2$.

Solution. Since a_1, a_2 are least elements of A , they belong to A . Using that a_1 is a least element of A , we obtain that $a_1 \leq a_2$. Further, using that a_2 is a least element of A , we obtain that $a_2 \leq a_1$. Combining these two inequalities, it follows that $a_1 = a_2$. $\square$



Remark

By [Exercise 2.1](#), a nonempty subset of $\mathbb{N}$ has at most one least element, to be called **the** least element, if it exists.

Well-ordering principle

Every nonempty subset of $\mathbb{N}$ has a least element.

Principle of mathematical induction

For each $n \in \mathbb{N}$, let $P(n)$ be a statement. Suppose $P(1)$ is true, and that whenever $P(n)$ holds for some $n \geq 1$, the statement $P(n+1)$ holds. Then $P(n)$ is true for all $n \geq 1$.

Principle of strong induction

For each $n \in \mathbb{N}$, let $P(n)$ be a statement. Suppose $P(1)$ is true, and that whenever $P(1), P(2), \dots, P(n)$ hold for some $n \geq 1$, the statement $P(n+1)$ holds. Then $P(n)$ is true for all $n \geq 1$.

i Remark

The above three principles are equivalent.

Exercise 2.2 (Infinitude of primes, by Saidak). Let $a_1 = 2$, and $a_{n+1} = a_n(a_n + 1)$ for any positive integer n . For any $n \in \mathbb{N}$, show that a_n has at least n distinct prime divisors.

Solution. For a positive integer n , let $P(n)$ denote the statement that $a_n \geq 1$ and a_n has at least n distinct prime divisors.

Since $a_1 = 2$, it follows that $P(1)$ is true.

Let n be a positive integer such that $P(n)$ holds. This gives that $a_n \geq 1$, and hence $a_n + 1 \geq 2$. Also note that the integers $a_n, a_n + 1$ have no common prime divisor, since any of their common divisors divides their difference, which is equal to 1. Hence, no prime divisor of a_n is a prime divisor of $a_n + 1$. By the induction hypothesis, a_n has at least n distinct prime divisors. Since $a_n + 1 \geq 2$, it admits at least one prime divisor. It follows that the product $a_n(a_n + 1)$ has at least $n + 1$ distinct prime divisors. Also note that $a_n(a_n + 1) \geq 2$ holds. This shows that $P(n+1)$ holds.

By induction, the statement $P(n)$ holds for all $n \in \mathbb{N}$. In particular, a_n has at least n distinct prime divisors. $\square$

Exercise 2.3 ().** Show that there are infinitely many prime numbers of the form $4m + 3$, where m is a nonnegative integer.

Solution. Let $a_1, a_2, \dots$ be a sequence of positive integers defined by

$$\begin{aligned} a_1 &:= 7, \\ a_{n+1} &:= a_n(4a_n + 3) \end{aligned}$$

for all $n \in \mathbb{N}$. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that a_n is a positive integer, 3 does not divide a_n , and a_n has at least n distinct prime divisors of the form $4m + 3$.

Since $a_1 = 7$, it follows that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Since a_k is a positive integer, it follows that $4a_k + 3$ is an odd positive integer ≥ 3 . Hence it has³ a prime divisor of the form $4m + 3$. Since 3 does not divide a_k , it follows⁴ that 3 does not divide $4a_k + 3$. Also note that any common factor of the integers a_k and $4a_k + 3$ divides

$$(4a_k + 3) - 4a_k,$$

³How does it follow? Prove it by induction that for any positive integer n , the product of any integers of the form $4m + 1$ is also of the form $4m + 1$.

⁴Why?

which is equal to 3. Since 3 does not divide a_k , we obtain that the integers a_k and $4a_k + 3$ have no common prime divisor. Hence, any prime divisor of $4a_k + 3$ of the form $4m + 3$ is different from the prime divisors of a_k . This shows that a_{k+1} is a positive integer, 3 does not divide a_{k+1} , and a_{k+1} has at least $k + 1$ distinct prime divisors of the form $4m + 3$. By induction, it follows that for any $n \in \mathbb{N}$, the integer a_n has n distinct prime divisors of the form $4m + 3$.

This shows that there are infinitely many prime numbers of the form $4m + 3$. $\square$

Exercise 2.4. Show that

$$2^{2^n} - 1$$

has at least n distinct prime divisors for all $n \in \mathbb{N}$.

Solution. For a positive integer n , let $P(n)$ denote the statement that $2^{2^n} - 1$ has at least n distinct prime divisors.

Since $2^{2^1} - 1 = 3$, it follows that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$2^{2^{k+1}} - 1 = (2^{2^k} - 1)(2^{2^k} + 1).$$

Also note that the integers $2^{2^k} - 1$ and $2^{2^k} + 1$ have no common prime divisors, since these are odd integers, and any of their common divisors divides their difference, which is equal to 2. By the induction hypothesis, the integer $2^{2^k} - 1$ has at least k distinct prime divisors. Since $2^{2^k} + 1 \geq 5$, it admits at least one prime divisor. It follows that the product $(2^{2^k} - 1)(2^{2^k} + 1)$ has at least $k + 1$ distinct prime divisors. This shows that $P(k + 1)$ holds.

By induction, the statement $P(n)$ holds for all $n \in \mathbb{N}$. $\square$

i Remark

Note that it is false that

$$2^{2^n} + 1$$

has at least n distinct prime divisors for all $n \in \mathbb{N}$. Indeed, the integer $2^{2^2} + 1 = 17$ is a prime.

Exercise 2.5. Let S be a finite set of size n . Show that the number of subsets of S is 2^n .

Solution. For a nonnegative integer n , let $P(n)$ denote the statement that if A is a set with n elements, then the number of subsets of A is 2^n .

Since the empty set has exactly one subset, namely itself, it follows that $P(0)$ holds.

Let k be a nonnegative integer such that $P(k)$ holds, that is, for **any** set with k elements, the number of its subsets is 2^k . We show that $P(k + 1)$ holds.

Let A be a set with $k + 1$ elements. Choose an element $a \in A$, and let $B = A \setminus \{a\}$. Note that B has exactly k elements. By the induction hypothesis, the number of subsets of B is 2^k . Any subset of A either contains a or does not contain a . The number of subsets of A that do not contain a is equal to the number of subsets of B , which is 2^k .

Note that any subset of A that contains a is of the form $C \cup \{a\}$, where C is a subset of B . Further, different subsets of B give rise to different subsets of A that contain a . That is, if C_1, C_2 are different subsets of B , then $C_1 \cup \{a\} \neq C_2 \cup \{a\}$. This shows that the number of subsets of A that contain a is also equal to the number of subsets of B , which is again 2^k . It follows that the total number of subsets of A is equal to

$$2^k + 2^k = 2^{k+1}.$$

This shows that $P(k + 1)$ holds.

By induction, it follows that for any nonnegative integer n , if A is a set of size n , then the number of subsets of A is 2^n . $\square$

Exercise 2.6. Show that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for all $n \in \mathbb{N}$.

Solution. For a positive integer n , let $P(n)$ denote the statement that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned} & \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \quad (\text{using } P(k)) \\ &= \frac{k(k+2) + 1}{(k+1)(k+2)} \\ &= \frac{(k+1)^2}{(k+1)(k+2)} \\ &= \frac{k+1}{k+2}. \end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n . $\square$

Exercise 2.7. Show that

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

for all $n \in \mathbb{N}$.

Solution. For a positive integer n , let $P(n)$ denote the statement that

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned} & \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2(k+1)-1)(2(k+1)+1)} \\ &= \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} \quad (\text{using } P(k)) \\ &= \frac{2k^2 + 3k + 1}{(2k+1)(2k+3)} \\ &= \frac{(k+1)(2k+1)}{(2k+1)(2k+3)} \\ &= \frac{k+1}{2(k+1)+1}. \end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n . $\square$

Exercise 2.8. Show that

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \cdots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

for all $n \in \mathbb{N}$.

Solution. For a positive integer n , let $P(n)$ denote the statement that

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \cdots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned} & \frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \cdots + \frac{1}{k(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} \\ &= \frac{k(k+3)}{4(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} \quad (\text{using } P(k)) \\ &= \frac{k(k+3)^2 + 4}{4(k+1)(k+2)(k+3)} \\ &= \frac{(k+1)(k+3)^2 - ((k+3)^2 - 4)}{4(k+1)(k+2)(k+3)} \\ &= \frac{(k+1)(k+3)^2 - ((k+3)^2 - 2^2)}{4(k+1)(k+2)(k+3)} \\ &= \frac{(k+1)(k+3)^2 - (k+1)(k+5)}{4(k+1)(k+2)(k+3)} \\ &= \frac{(k+1)((k+3)^2 - (k+5))}{4(k+1)(k+2)(k+3)} \\ &= \frac{k^2 + 6k + 9 - (k+5)}{4(k+2)(k+3)} \\ &= \frac{k^2 + 5k + 4}{4(k+2)(k+3)} \\ &= \frac{(k+1)((k+1)+3)}{4((k+1)+1)((k+1)+2)}. \end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n . $\square$

Exercise 2.9. Show that

$$3 + 11 + \cdots + (8n-5) = 4n^2 - n$$

for all $n \in \mathbb{N}$.

Solution. For a positive integer n , let $P(n)$ denote the statement that

$$3 + 11 + \cdots + (8n-5) = 4n^2 - n.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
& 3 + 11 + \cdots + (8k - 5) + (8(k + 1) - 5) \\
&= (4k^2 - k) + (8k + 3) \quad (\text{using } P(k)) \\
&= 4k^2 + 7k + 3 \\
&= 4k^2 + 8k + 4 - (k + 1) \\
&= 4(k + 1)^2 - (k + 1).
\end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k + 1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n . $\square$

Exercise 2.10. Show that

$$1^2 + 3^2 + \cdots + (2n - 1)^2 = \frac{4n^3 - n}{3}$$

for all $n \in \mathbb{N}$.

Solution. For a positive integer n , let $P(n)$ denote the statement that

$$1^2 + 3^2 + \cdots + (2n - 1)^2 = \frac{4n^3 - n}{3}.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
& 1^2 + 3^2 + \cdots + (2k - 1)^2 + (2(k + 1) - 1)^2 \\
&= \frac{4k^3 - k}{3} + (2k + 1)^2 \quad (\text{using } P(k)) \\
&= \frac{4k^3 - k + 3(2k + 1)^2}{3} \\
&= \frac{4k^3 - k + 12k^2 + 12k + 3}{3} \\
&= \frac{4k^3 + 12k^2 + 11k + 3}{3} \\
&= \frac{4(k + 1)^3 - (k + 1)}{3}.
\end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k + 1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n . $\square$

Exercise 2.11. Show that

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1}n^2 = (-1)^{n+1} \frac{n(n + 1)}{2}$$

for all $n \in \mathbb{N}$.

Solution. For a positive integer n , let $P(n)$ denote the statement that

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1}n^2 = (-1)^{n+1} \frac{n(n + 1)}{2}.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
& 1^2 - 2^2 + 3^2 - 4^2 + \dots + (-1)^{k+1}k^2 + (-1)^{(k+1)+1}(k+1)^2 \\
&= (-1)^{k+1}\frac{k(k+1)}{2} + (-1)^{k+2}(k+1)^2 \quad (\text{using } P(k)) \\
&= (-1)^{k+1}\left(\frac{k(k+1)}{2} - (k+1)^2\right) \\
&= (-1)^{k+1}\left(\frac{k(k+1) - 2(k+1)^2}{2}\right) \\
&= (-1)^{k+1}\left(\frac{-(k+1)(k+2)}{2}\right) \\
&= (-1)^{(k+1)+1}\frac{(k+1)((k+1)+1)}{2}.
\end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n . $\square$

Exercise 2.12. Show that

$$\begin{aligned}
1^2 + 2^2 + 3^2 + \dots + n^2 &= \frac{n(n+1)(2n+1)}{6}, \\
1^3 + 2^3 + 3^3 + \dots + n^3 &= \left(\frac{n(n+1)}{2}\right)^2
\end{aligned}$$

hold for any positive integer n .

Solution. For a positive integer n , let $P(n)$ denote the statement that

$$\begin{aligned}
1^2 + 2^2 + 3^2 + \dots + n^2 &= \frac{n(n+1)(2n+1)}{6}, \\
1^3 + 2^3 + 3^3 + \dots + n^3 &= \left(\frac{n(n+1)}{2}\right)^2.
\end{aligned}$$

Note that $P(1)$ holds. Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
& 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 \\
&= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \quad (\text{using } P(k)) \\
&= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \\
&= \frac{(k+1)(k(2k+1) + 6(k+1))}{6} \\
&= \frac{(k+1)(2k^2 + k + 6k + 6)}{6} \\
&= \frac{(k+1)(2k^2 + 7k + 6)}{6} \\
&= \frac{(k+1)(k+2)(2k+3)}{6} \\
&= \frac{(k+1)(k+2)(2(k+1)+1)}{6}.
\end{aligned}$$

Also note that

$$\begin{aligned}
& 1^3 + 2^3 + 3^3 + \cdots + k^3 + (k+1)^3 \\
&= \left(\frac{k(k+1)}{2} \right)^2 + (k+1)^3 \quad (\text{using } P(k)) \\
&= (k+1)^2 \left(\frac{k^2}{4} + (k+1) \right) \\
&= (k+1)^2 \left(\frac{k^2}{4} + k + 1 \right) \\
&= (k+1)^2 \left(\frac{(k+2)^2}{4} \right) \\
&= \left(\frac{(k+1)((k+1)+1)}{2} \right)^2.
\end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n . □

Exercise 2.13. Show that

$$\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n-1} - \frac{1}{2n}$$

for all $n \in \mathbb{N}$.

Solution. For a positive integer n , let $P(n)$ denote the statement that

$$\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n-1} - \frac{1}{2n}.$$

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
& \frac{1}{(k+1)+1} + \cdots + \frac{1}{2k+1} + \frac{1}{2(k+1)} \\
&= \frac{1}{k+1} + \frac{1}{(k+1)+1} + \cdots + \frac{1}{2k} + \frac{1}{2k+1} + \frac{1}{2(k+1)} - \frac{1}{k+1} \\
&= \frac{1}{k+1} + \frac{1}{(k+1)+1} + \cdots + \frac{1}{2k} + \frac{1}{2k+1} - \frac{1}{2(k+1)} \\
&= \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2k-1} - \frac{1}{2k} \right) + \frac{1}{2k+1} - \frac{1}{2(k+1)} \\
& \quad \quad \quad (\text{using } P(k)) \\
&= \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2k-1} - \frac{1}{2k} \right) + \frac{1}{2k+1} - \frac{1}{2k+2} \\
&= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{(2(k+1))-1} - \frac{1}{(2(k+1))}.
\end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n . □

**Definition**

An integer is called **divisible** by a nonzero integer m if it is equal to mk for some integer k .

Exercise 2.14. Show that $n^3 + 5n$ is divisible by 6 for all $n \in \mathbb{N}$.

Solution. For a positive integer n , let $P(n)$ denote the statement that $n^3 + 5n$ is divisible by 6.

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}(k+1)^3 + 5(k+1) &= (k^3 + 3k^2 + 3k + 1) + (5k + 5) \\ &= (k^3 + 5k) + (3k^2 + 3k + 6).\end{aligned}$$

By the induction hypothesis, the integer $k^3 + 5k$ is divisible by 6. Also note that $3k^2 + 3k + 6 = 3(k^2 + k + 2)$ is divisible by 3. Further, one of the integers $k, k+1$ is even, and hence $k^2 + k$ is even. This gives that $k^2 + k + 2$ is even, and hence $3(k^2 + k + 2)$ is divisible by 2. It follows that $3(k^2 + k + 2)$ is divisible by 6. This shows that $(k+1)^3 + 5(k+1)$ is divisible by 6. This proves that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds.

By the principle of induction, it follows that $P(n)$ holds for any positive integer n . $\square$

Exercise 2.15. Show that $5^{2n} - 1$ is divisible by 8 for all $n \in \mathbb{N}$.

Solution. For a positive integer n , let $P(n)$ denote the statement that $5^{2n} - 1$ is divisible by 8.

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}5^{2(k+1)} - 1 &= 5^{2k+2} - 1 \\ &= 5^2 \cdot 5^{2k} - 1 \\ &= 25 \cdot 5^{2k} - 1 \\ &= (24 + 1) \cdot 5^{2k} - 1 \\ &= 24 \cdot 5^{2k} + (5^{2k} - 1).\end{aligned}$$

By the induction hypothesis, the integer $5^{2k} - 1$ is divisible by 8. Also note that $24 \cdot 5^{2k}$ is divisible by 8. This shows that $5^{2(k+1)} - 1$ is divisible by 8. This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds.

By the principle of induction, it follows that $P(n)$ holds for any positive integer n . $\square$

Exercise 2.16. Show that $5^n - 4n - 1$ is divisible by 16 for all $n \in \mathbb{N}$.

Solution. For a positive integer n , let $P(n)$ denote the statement that $5^n - 4n - 1$ is divisible by 16.

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}5^{k+1} - 4(k+1) - 1 &= 5 \cdot 5^k - 4k - 4 - 1 \\ &= 5 \cdot (5^k - 4k - 1) + 5 \cdot (4k + 1) - 4k - 5 \\ &= 5 \cdot (5^k - 4k - 1) + 16k.\end{aligned}$$

By the induction hypothesis, the integer $5^k - 4k - 1$ is divisible by 16. Using the above, it follows that $5^{k+1} - 4(k+1) - 1$ is divisible by 16. This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds.

By the principle of induction, it follows that $P(n)$ holds for any positive integer n . $\square$

Exercise 2.17. Show that $6^n - 5n - 1$ is divisible by 25 for all $n \in \mathbb{N}$.

Exercise 2.18. Show that $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9 for all $n \in \mathbb{N}$.

Solution. For a positive integer n , let $P(n)$ denote the statement that $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9.

Note that $P(1)$ holds.

Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned} & (k+1)^3 + (k+2)^3 + (k+3)^3 \\ &= (k^3 + (k+1)^3 + (k+2)^3) - k^3 + (k+3)^3 \\ &= (k^3 + (k+1)^3 + (k+2)^3) - k^3 + (k^3 + 9k^2 + 27k + 27) \\ &= (k^3 + (k+1)^3 + (k+2)^3) + (9k^2 + 27k + 27). \end{aligned}$$

By the induction hypothesis, the integer $k^3 + (k+1)^3 + (k+2)^3$ is divisible by 9. Also note that $9k^2 + 27k + 27$ is divisible by 9. This shows that $(k+1)^3 + (k+2)^3 + (k+3)^3$ is divisible by 9. This implies that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds.

By the principle of induction, it follows that $P(n)$ holds for any positive integer n . $\square$

Exercise 2.19. Show that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

for all $n \in \mathbb{N}$ satisfying $n > 1$.

Solution. For a positive integer n , let $P(n)$ denote the statement that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

holds.

Note that $P(2)$ holds, since

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} = 1 + \frac{\sqrt{2}}{2} > \sqrt{2}.$$

Let k be a positive integer such that $k \geq 2$ and $P(k)$ holds. Note that

$$\begin{aligned} & \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \\ & > \sqrt{k} + \frac{1}{\sqrt{k+1}} \quad (\text{using } P(k)) \\ & \geq \sqrt{k+1}, \end{aligned}$$

where the last inequality follows since

$$\begin{aligned} k - \left(\sqrt{k+1} - \frac{1}{\sqrt{k+1}} \right)^2 &= k - (k+1) + 2 - \frac{1}{k+1} \\ &= 1 - \frac{1}{k+1} \\ &> 0. \end{aligned}$$

This shows that if k is a positive integer satisfying $k \geq 2$ and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n satisfying $n \geq 2$. $\square$

Exercise 2.20. Show that $3^n \geq n^2$ for all $n \in \mathbb{N}$.

Solution. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that $3^n \geq n^2$. Note that $P(1)$ holds. Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}(k+1)^2 &= k^2 + 2k + 1 \\ &\leq 3^k + 2k + 1 \quad (\text{using } P(k)) \\ &\leq 3^k + 3^k + 3^k \quad (\text{using } k \geq 1) \\ &= 3 \cdot 3^k \\ &= 3^{k+1}.\end{aligned}$$

This shows that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n . $\square$

Exercise 2.21. Show that $n! > 2^n$ for all $n \in \mathbb{N}$ satisfying $n \geq 4$.

Solution. For a positive integer n , let $P(n)$ denote the statement that $n! > 2^n$.

Note that $P(4)$ holds, since $4! = 24 > 16 = 2^4$.

Let k be a positive integer such that $k \geq 4$ and $P(k)$ holds. Note that

$$\begin{aligned}(k+1)! &= (k+1)k! \\ &\geq (k+1) \cdot 2^k \quad (\text{using } P(k)) \\ &> 2 \cdot 2^k \quad (\text{using } k \geq 4) \\ &= 2^{k+1}.\end{aligned}$$

This shows that if k is a positive integer satisfying $k \geq 4$ and $P(k)$ holds, then $P(k+1)$ also holds.

By the principle of induction, it follows that $P(n)$ holds for any positive integer n satisfying $n \geq 4$. $\square$

Exercise 2.22 ().** Show that

$$\left(1 + \frac{1}{1^3}\right)\left(1 + \frac{1}{2^3}\right)\left(1 + \frac{1}{3^3}\right)\cdots\left(1 + \frac{1}{n^3}\right) < 3$$

holds for all $n \in \mathbb{N}$.

i Remark ()**

Does taking $P(n)$ to be the statement that

$$\left(1 + \frac{1}{1^3}\right)\left(1 + \frac{1}{2^3}\right)\left(1 + \frac{1}{3^3}\right)\cdots\left(1 + \frac{1}{n^3}\right) < 3$$

help?

Solution. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that

$$\left(1 + \frac{1}{1^3}\right)\left(1 + \frac{1}{2^3}\right)\left(1 + \frac{1}{3^3}\right)\cdots\left(1 + \frac{1}{n^3}\right) \leq 3 - \frac{1}{n}.$$

Note that $P(1)$ holds. Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
& \left(1 + \frac{1}{1^3}\right) \left(1 + \frac{1}{2^3}\right) \left(1 + \frac{1}{3^3}\right) \cdots \left(1 + \frac{1}{k^3}\right) \left(1 + \frac{1}{(k+1)^3}\right) \\
& \leq \left(3 - \frac{1}{k}\right) \left(1 + \frac{1}{(k+1)^3}\right) \\
& = 3 - \frac{1}{k} + \frac{3}{(k+1)^3} - \frac{1}{k(k+1)^3} \\
& = 3 - \frac{1}{k+1} + \frac{1}{k+1} - \frac{1}{k} + \frac{3}{(k+1)^3} - \frac{1}{k(k+1)^3} \\
& = 3 - \frac{1}{k+1} - \frac{1}{k(k+1)} + \frac{3}{(k+1)^3} - \frac{1}{k(k+1)^3} \\
& = 3 - \frac{1}{k+1} - \frac{(k+1)^2 - 3k + 1}{k(k+1)^3} \\
& = 3 - \frac{1}{k+1} - \frac{k^2 - k + 2}{k(k+1)^3} \\
& = 3 - \frac{1}{k+1} - \frac{k(k-1) + 2}{k(k+1)^3} \\
& < 3 - \frac{1}{k+1}
\end{aligned}$$

holds. This proves that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n . This implies the given inequality for all $n \in \mathbb{N}$.

□

Exercise 2.23 ().** Show that

$$1 + \frac{1}{2^3} + \frac{1}{3^3} + \cdots + \frac{1}{n^3} < \frac{3}{2}$$

for all $n \in \mathbb{N}$.

Solution. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that

$$1 + \frac{1}{2^3} + \frac{1}{3^3} + \cdots + \frac{1}{n^3} \leq \frac{3}{2} - \frac{1}{2n^2}.$$

Note that $P(1)$ holds. Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
& 1 + \frac{1}{2^3} + \frac{1}{3^3} + \cdots + \frac{1}{k^3} + \frac{1}{(k+1)^3} \\
& \leq \left(\frac{3}{2} - \frac{1}{2k^2} \right) + \frac{1}{(k+1)^3} \quad (\text{using } P(k)) \\
& = \frac{3}{2} - \frac{1}{2k^2} + \frac{1}{(k+1)^3} \\
& = \frac{3}{2} - \frac{1}{2(k+1)^2} + \frac{1}{2(k+1)^2} - \frac{1}{2k^2} + \frac{1}{(k+1)^3} \\
& = \frac{3}{2} - \frac{1}{2(k+1)^2} - \frac{(k+1)^3 - k^2(k+1) - 2k^2}{2k^2(k+1)^3} \\
& = \frac{3}{2} - \frac{1}{2(k+1)^2} - \frac{k^3 + 3k^2 + 3k + 1 - k^3 - k^2 - 2k^2}{2k^2(k+1)^3} \\
& = \frac{3}{2} - \frac{1}{2(k+1)^2} - \frac{3k+1}{2k^2(k+1)^3} \\
& < \frac{3}{2} - \frac{1}{2(k+1)^2}.
\end{aligned}$$

This proves that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n . This implies the given inequality for all $n \in \mathbb{N}$. □

Exercise 2.24 ().** Show that

$$\left(1 + \frac{1}{2}\right) \left(1 + \frac{1}{2^2}\right) \left(1 + \frac{1}{2^3}\right) \cdots \left(1 + \frac{1}{2^n}\right) < \frac{5}{2}$$

holds for all $n \in \mathbb{N}$.

Solution. For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that

$$\left(1 + \frac{1}{2}\right) \left(1 + \frac{1}{2^2}\right) \left(1 + \frac{1}{2^3}\right) \cdots \left(1 + \frac{1}{2^n}\right) \leq \frac{5}{2} \left(1 - \frac{2}{2^{n+1} + 1}\right).$$

Note that $P(1)$ holds. Let k be a positive integer such that $P(k)$ holds. Note that

$$\begin{aligned}
& \left(1 + \frac{1}{2}\right) \left(1 + \frac{1}{2^2}\right) \left(1 + \frac{1}{2^3}\right) \cdots \left(1 + \frac{1}{2^k}\right) \left(1 + \frac{1}{2^{k+1}}\right) \\
& \leq \frac{5}{2} \left(1 - \frac{2}{2^{k+1} + 1}\right) \left(1 + \frac{1}{2^{k+1}}\right) \\
& \leq \frac{5}{2} \times \frac{2^{k+1} - 1}{2^{k+1}} \\
& \leq \frac{5}{2} \left(1 - \frac{1}{2^{k+1}}\right) \\
& \leq \frac{5}{2} \left(1 - \frac{1}{2^{k+2} + 1}\right).
\end{aligned}$$

This proves that if k is a positive integer and $P(k)$ holds, then $P(k+1)$ also holds. By the principle of induction, it follows that $P(n)$ holds for any positive integer n . This implies the given inequality for all $n \in \mathbb{N}$. □

 Warning

It is important to verify the base case, that is, verifying that $P(1)$ holds, even if it is trivial. For example, if we define $P(n)$ to be the statement that

$$n = n + 1,$$

then $P(n)$ implies $P(n + 1)$ for any $n \in \mathbb{N}$. However, $P(1)$ does not hold. Another example is to take $Q(n)$ to be the statement that $2n + 1$ is even, then $Q(n)$ implies $Q(n + 1)$ for any $n \in \mathbb{N}$. However, $Q(1)$ does not hold.

Here are a few exercises where strong induction is useful.

Exercise 2.25. Show that every positive integer greater than 1 is a prime or is a product of prime numbers.

Exercise 2.26. Let the integers $x_1, x_2, \dots$ be defined by

$$x_1 := 1,$$

$$x_2 := 2,$$

$$x_{n+2} := \frac{1}{2}(x_n + x_{n+1})$$

for all $n \in \mathbb{N}$. Show that $1 \leq x_n \leq 2$ for all $n \in \mathbb{N}$.

 Tip

For any $n \in \mathbb{N}$, let $P(n)$ denote the statement that $1 \leq x_m \leq 2$ for all $m \in \mathbb{N}$ satisfying $m \leq n$.

Exercise 2.27. The Fibonacci sequence $F_0, F_1, F_2, \dots$ is defined by

$$F_0 := 0,$$

$$F_1 := 1,$$

$$F_2 := 1,$$

$$F_{n+2} := F_n + F_{n+1}$$

for all $n \in \mathbb{N}$. Show that

$$F_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right)$$

for all $n \in \mathbb{N} \cup \{0\}$.

Exercise 2.28. Show that

$$F_0^2 + F_1^2 + F_2^2 + \dots + F_n^2 = F_n F_{n+1}$$

for all $n \in \mathbb{N} \cup \{0\}$.

Linear algebra

Chapter 3. Matrices

§3.1 Vectors

§3.1.1 Definitions and examples



A vector is a column of numbers. The numbers in the vector are called its **components** or **coordinates**.



Remark

Throughout this course, most of the vectors to be considered, will have real or complex components only.

Recall that for two sets A, B , their cartesian product $A \times B$ is defined as

$$A \times B := \{(a, b) \mid a \in A, b \in B\}.$$

The cartesian product of $\mathbb{R}$ with itself is denoted by $\mathbb{R}^2$. In other words,

$$\mathbb{R}^2 := \{(x, y) \mid x \in \mathbb{R}, y \in \mathbb{R}\}.$$

Similarly, one defines $\mathbb{R}^3$ as $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$, that is,

$$\mathbb{R}^3 := \{(x, y, z) \mid x \in \mathbb{R}, y \in \mathbb{R}, z \in \mathbb{R}\}.$$



Definition

If $n \geq 2$ is an integer and $A_1, A_2, \dots, A_n$ are sets, then their **cartesian product** $A_1 \times A_2 \times \dots \times A_n$ is defined as

$$A_1 \times A_2 \times \dots \times A_n := \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for all } i = 1, 2, \dots, n\}.$$

The n -fold cartesian product of $\mathbb{R}$ with itself, that is, the cartesian product of n copies of $\mathbb{R}$, is denoted by $\mathbb{R}^n$. In other words,

$$\mathbb{R}^n := \{(x_1, x_2, \dots, x_n) \mid x_i \in \mathbb{R} \text{ for all } i = 1, 2, \dots, n\}.$$

Similarly, one defines $\mathbb{C}^n$ as the n -fold cartesian product of $\mathbb{C}$ with itself, that is,

$$\mathbb{C}^n := \{(z_1, z_2, \dots, z_n) \mid z_i \in \mathbb{C} \text{ for all } i = 1, 2, \dots, n\}.$$

**Example**

The elements

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

of $\mathbb{R}^2$ are vectors. The elements

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 10 \\ 20 \\ -2025 \end{pmatrix}$$

of $\mathbb{R}^3$ are vectors too. The elements

$$\begin{pmatrix} 1+i \\ 2-i \\ -3i \end{pmatrix}, \begin{pmatrix} 0 \\ \pi \\ 1+i \end{pmatrix}$$

of $\mathbb{C}^3$ are vectors as well.

§3.1.2 Vector operations**Definition**

Two vectors $\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ and $\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$ in $\mathbb{R}^n$ are added component-wise. That is, the **sum** of two elements $\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ and $\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$ in $\mathbb{R}^n$ is defined as

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} := \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}.$$

If $c \in \mathbb{R}$ is a real number and $v = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ is a vector in $\mathbb{R}^n$, then the **scalar multiplication** of c with the vector v is defined as

$$c \cdot \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} := \begin{pmatrix} c \cdot x_1 \\ c \cdot x_2 \\ \vdots \\ c \cdot x_n \end{pmatrix}.$$

Note that $1 \cdot v$ is equal to v . Instead of $(-1) \cdot v$, we write $-v$. Further, $c \cdot v$ is often written as cv .

**Warning**

In the above definition, we have used the same notation “+” for addition of two vectors and addition of two real numbers, and the same notation “.” for scalar multiplication of a vector by a real number and multiplication of two real numbers. The meaning will be clear from the context.

 Remark

Note that the sum of two vectors in $\mathbb{R}^n$ is again a vector in $\mathbb{R}^n$. Also, the scalar multiplication of a vector in $\mathbb{R}^n$ by a real number is again a vector in $\mathbb{R}^n$.

 Remark

Similar definitions work for vectors in $\mathbb{C}^n$.

 Example

Consider the vectors $u = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $v = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$ in $\mathbb{R}^3$. Note that

$$u + v = \begin{pmatrix} 1+4 \\ 2+5 \\ 3+6 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \\ 9 \end{pmatrix},$$

and if $c = 3$, then

$$c \cdot u = 3 \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \cdot 1 \\ 3 \cdot 2 \\ 3 \cdot 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix}.$$

§3.2 Matrix

§3.2.1 Definitions and examples

 Definition

A **matrix** is a rectangular array of numbers arranged in rows and columns. A matrix with m rows and n columns is called an $m \times n$ **matrix**. The numbers in the matrix are called its **entries** or **elements**. A matrix is called a **square matrix** if its number of rows, and its number of columns are equal.

 Definition

Two matrices are **equal** if they have the same size and their corresponding entries are equal.

 Remark

The **size** of a matrix is always given in the form “(the number of rows) $\times$ (the number of columns)”.

**Example**

The following is a 2×3 matrix:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}.$$

The following is a 3×2 matrix:

$$B = \begin{pmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{pmatrix}.$$

If A is a matrix, its (i, j) -th entry, that is, the entry common to the i -th row and the j -th column is denoted by a_{ij} . Thus an $m \times n$ matrix A can be represented as

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}.$$

One also writes $A = (a_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}$, to denote the matrix A , or one writes $A = (a_{ij})$ if the size of A is clear from the context.

i Remark

Note that for an $m \times n$ matrix A , one writes $A = (\text{the } (i, j)\text{-th entry of } A)_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}$.

If $A = (a_{ij})$ is an $n \times n$ matrix, then the entries $a_{11}, a_{22}, \dots, a_{nn}$ are called its **diagonal entries**. A square matrix is called a **diagonal matrix** if its entries, other the diagonal entries, are equal to zero.

§3.2.2 Matrix times a vector

The product

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$$

is defined by “combining” each row of the matrix with the vector as follows:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} = \begin{pmatrix} 1 \times 7 + 2 \times 8 + 3 \times 9 \\ 4 \times 7 + 5 \times 8 + 6 \times 9 \end{pmatrix},$$

which is equal to the vector $\begin{pmatrix} 50 \\ 122 \end{pmatrix}$.

As we see, for this to work, the number of columns of the matrix must be equal to the number of rows of the vector. Thus, when the product of an $m \times n$ matrix with a vector in $\mathbb{R}^n$ is defined similarly, the result is a vector in $\mathbb{R}^m$.

 Definition

If

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

is an $m \times n$ matrix with entries in $\mathbb{R}$ and $\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ is a vector in $\mathbb{R}^n$, then one defines

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} := \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{pmatrix}.$$

Note that the result is a vector in $\mathbb{R}^m$.

§3.2.3 Matrix times a matrix

Similar to the case of vectors, one defines addition, subtraction, and scalar multiplication of matrices entry-wise. For addition and subtraction, the two matrices must be of the same size. Indeed, if

$$A = (a_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}, B = (b_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}$$

are two $m \times n$ matrices, then their **sum** $A + B$ and **difference** $A - B$ are defined as

$$A + B := (a_{ij} + b_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}$$

and

$$A - B := (a_{ij} - b_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}$$

respectively. If c is a real number and $A = (a_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}$ is an $m \times n$ matrix, then the **scalar multiplication** of A with c is defined as

$$c \cdot A := (c \cdot a_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}.$$

Note that $1 \cdot A$ is equal to A . Instead of $-1 \cdot A$, we write $-A$. Further, $c \cdot A$ is often written as cA .

Remark

Note that the sum of two matrices with entries in $\mathbb{R}$ is a matrix with entries in $\mathbb{R}$. Also, the scalar multiplication of a matrix with entries in $\mathbb{R}$ by a real number is again a matrix with entries in $\mathbb{R}$.

Remark

Similar definitions work for matrices with entries in $\mathbb{C}$.

**Example**

If

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

and

$$B = \begin{pmatrix} 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix},$$

then

$$A + B = \begin{pmatrix} 1+7 & 2+8 & 3+9 \\ 4+10 & 5+11 & 6+12 \end{pmatrix} = \begin{pmatrix} 8 & 10 & 12 \\ 14 & 16 & 18 \end{pmatrix},$$

and if $c = 3$, then

$$c \cdot A = 3 \cdot \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 3 \cdot 1 & 3 \cdot 2 & 3 \cdot 3 \\ 3 \cdot 4 & 3 \cdot 5 & 3 \cdot 6 \end{pmatrix} = \begin{pmatrix} 3 & 6 & 9 \\ 12 & 15 & 18 \end{pmatrix}.$$

Exercise 3.1. If A, B, C are matrices of the same size, show that

$$A + (B + C) = (A + B) + C.$$

Exercise 3.2. If A, B are matrices of the same size, show that

$$A + B = B + A.$$

Exercise 3.3. If A is an $m \times n$ matrix, show that

$$A + 0_{m \times n} = 0_{m \times n} + A = A$$

holds, where $0_{m \times n}$ denotes the $m \times n$ zero matrix, that is, the $m \times n$ matrix whose entries are all zero.

Exercise 3.4. Show that if A is a matrix and c, d are real numbers, then

$$(c + d)A = cA + dA$$

and

$$c(dA) = (cd)A.$$

 Definition

The **transpose** of an $m \times n$ matrix

$$A = (a_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}$$

is the $n \times m$ matrix

$$A^T := (a_{ji})_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq m}}$$

That is, if

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix},$$

then its **transpose** is the $n \times m$ matrix

$$A^T := \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}.$$

 Example

If

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix},$$

then its transpose is the 3×2 matrix

$$A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

 Remark

Note that the transpose of a matrix is obtained by interchanging its rows and columns.

Exercise 3.5. If A is a matrix, what is $(A^T)^T$?

Exercise 3.6. Show that if A, B are matrices of the same size, then

$$(A + B)^T = A^T + B^T.$$

Solution. Write

$$A = (a_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}, B = (b_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}.$$

Note that

$$\begin{aligned}
(A+B)^T &= \left((a_{ij} + b_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}} \right)^T \\
&= (a_{ji} + b_{ji})_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq m}} \\
&= (a_{ji})_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq m}} + (b_{ji})_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq m}} \\
&= A^T + B^T.
\end{aligned}$$

□

Exercise 3.7. Show that if A denotes a matrix, then

$$(A^T)^T = A.$$

Solution. Write

$$A = (a_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}, B = (b_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}.$$

Note that

$$\begin{aligned}
(A^T)^T &= \left((a_{ji})_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq m}} \right)^T \\
&= (a_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}} \\
&= A.
\end{aligned}$$

□

Exercise 3.8. Show that if A, B are matrices, then

$$(AB)^T = B^T A^T.$$

Solution. Suppose A is an $m \times n$ matrix and B is an $n \times p$ matrix. Write

$$A = (a_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq n}}, B = (b_{ij})_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq p}}.$$

Note that

$$\begin{aligned}
(AB)^T &= \left((c_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq p}} \right)^T \\
&= (c_{ji})_{\substack{1 \leq i \leq p, \\ 1 \leq j \leq m}},
\end{aligned}$$

where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

for all $1 \leq i \leq m$ and $1 \leq j \leq p$. Also note that

$$\begin{aligned}
B^T A^T &= (b_{ji})_{\substack{1 \leq i \leq p, \\ 1 \leq j \leq n}} (a_{ji})_{\substack{1 \leq i \leq n, \\ 1 \leq j \leq m}} \\
&= (d_{ij})_{\substack{1 \leq i \leq p, \\ 1 \leq j \leq m}},
\end{aligned}$$

where

$$d_{ij} = b_{i1}a_{1j} + b_{i2}a_{2j} + \cdots + b_{in}a_{nj}$$

for all $1 \leq i \leq p$ and $1 \leq j \leq m$. Since

$$c_{ji} = a_{j1}b_{1i} + a_{j2}b_{2i} + \cdots + a_{jn}b_{ni} = b_{i1}a_{1j} + b_{i2}a_{2j} + \cdots + b_{in}a_{nj} = d_{ij}$$

for all $1 \leq i \leq p$ and $1 \leq j \leq m$, we have

$$(AB)^T = B^T A^T.$$

□



Definition

The **product** of an $m \times n$ matrix A with an $n \times p$ matrix B is defined as follows:

$$AB := (c_{ij})_{\substack{1 \leq i \leq m, \\ 1 \leq j \leq p}},$$

where

$$c_{ij} := a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

for all $1 \leq i \leq m$ and $1 \leq j \leq p$. In other words, given an $m \times n$ matrix

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

and an $n \times p$ matrix

$$B = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{pmatrix},$$

we have

$$AB = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & c_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mp} \end{pmatrix},$$

where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

for all $1 \leq i \leq m$ and $1 \leq j \leq p$. Note that the result is an $m \times p$ matrix.

In other words, the (i, j) -th entry of the product AB is obtained by “combining” the i -th row of A with the j -th column of B .



Remark

Note that the product AB is defined only when the number of columns of A is equal to the number of rows of B .

**Example**

If

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

and

$$B = \begin{pmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{pmatrix},$$

then the product AB is the 2×2 matrix

$$\begin{aligned} AB &= \begin{pmatrix} 1 \times 7 + 2 \times 9 + 3 \times 11 & 1 \times 8 + 2 \times 10 + 3 \times 12 \\ 4 \times 7 + 5 \times 9 + 6 \times 11 & 4 \times 8 + 5 \times 10 + 6 \times 12 \end{pmatrix} \\ &= \begin{pmatrix} 58 & 64 \\ 139 & 154 \end{pmatrix}. \end{aligned}$$

Exercise 3.9. Show that if A, B, C are matrices such that the products $A(BC)$ and $(AB)C$ are defined, then

$$A(BC) = (AB)C.$$

Exercise 3.10. Provide examples to show that in general, matrix multiplication is not commutative, that is, $AB \neq BA$ for some matrices A, B .

Exercise 3.11 ().** Show that for any $n \in \mathbb{N}$ with $n \geq 2$, there are $n \times n$ matrices A, B such that $AB \neq BA$.

**Question**

Can induction be used for the above exercise?

Exercise 3.12. If A is an $n \times n$ matrix, show that

$$AI_n = I_n A = A$$

holds, where I_n denotes the $n \times n$ diagonal matrix, with all diagonal entries equal to 1, that is,

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}.$$

**Remark**

The matrix I_n as above, is called the **identity matrix** of order n , or the $n \times n$ **identity matrix**.

Exercise 3.13. If A is an $m \times n$ matrix, show that

$$I_m A = A, AI_n = A$$

where I_m (resp. I_n) denote the $m \times m$ (resp. $n \times n$) identity matrix.

Exercise 3.14. If A is an $m \times n$ matrix with entries in $\mathbb{R}$ and c is a real number, then

$$cA = (cI_n)A = A(cI_m),$$

where I_n (resp. I_m) denotes the $n \times n$ (resp. $m \times m$) identity matrix.

i Remark

Given a square matrix A , that is, a matrix with the same number of rows and columns, the product AA is often denoted by A^2 . Similarly, one defines $A^3, A^4, \dots$

Exercise 3.15. Show that if

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

then

$$A^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}$$

for all $n \in \mathbb{N}$ with $n \geq 1$, where F_n denotes the n -th Fibonacci number.

Exercise 3.16. Compute

$$\begin{pmatrix} 0 & \cdots & 0 & 1 \\ 0 & \cdots & 1 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 1 & \cdots & 0 & 0 \end{pmatrix}^2.$$

 Definition

Let A be an $n \times n$ matrix, and k be a positive integer. The **k -th power** of A , denoted by A^k , is defined as the product of k copies of A .

? Question

Suppose A, B are matrices.

- Under what conditions are the products AB and BA defined?
- Under what conditions is the product $B^T A^T$ defined?

Exercise 3.17. Show that if A, B, C are matrices such that the sum $B + C$, and the products $AB, AC, A(B + C)$ are defined, then

$$A(B + C) = AB + AC.$$

Exercise 3.18. Show that if A, B, C are matrices such that the sum $B + C$, and the products $AB, AC, A(B + C)$ are defined, then

$$(A + B)C = AC + BC.$$

§3.2.4 Few to many more!



Exercise

Suppose A is a 3×2 matrix, and

$$A \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 8 \\ 14 \end{pmatrix},$$

$$A \begin{pmatrix} 3 \\ 7 \end{pmatrix} = \begin{pmatrix} -4 \\ 8 \\ 20 \end{pmatrix}.$$

Determine

$$A \begin{pmatrix} 103 \\ 407 \end{pmatrix}, A \begin{pmatrix} 97 \\ 393 \end{pmatrix}.$$

Solution. Observe that

$$\begin{pmatrix} 103 \\ 407 \end{pmatrix} = 100 \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \begin{pmatrix} 3 \\ 7 \end{pmatrix}.$$

This yields

$$\begin{aligned} A \begin{pmatrix} 103 \\ 407 \end{pmatrix} &= A \left(100 \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right) \\ &= 100 A \begin{pmatrix} 1 \\ 4 \end{pmatrix} + A \begin{pmatrix} 3 \\ 7 \end{pmatrix} \\ &= 100 \begin{pmatrix} 2 \\ 8 \\ 14 \end{pmatrix} + \begin{pmatrix} -4 \\ 8 \\ 20 \end{pmatrix} \\ &= \begin{pmatrix} 200 - 4 \\ 800 + 8 \\ 1400 + 20 \end{pmatrix} \\ &= \begin{pmatrix} 196 \\ 808 \\ 1420 \end{pmatrix}. \end{aligned}$$

Also note that

$$\begin{pmatrix} 97 \\ 393 \end{pmatrix} = 100 \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 7 \end{pmatrix}.$$

Using this we obtain

$$\begin{aligned}
A \begin{pmatrix} 97 \\ 393 \end{pmatrix} &= A \left(100 \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right) \\
&= 100A \begin{pmatrix} 1 \\ 4 \end{pmatrix} - A \begin{pmatrix} 3 \\ 7 \end{pmatrix} \\
&= 100 \begin{pmatrix} 2 \\ 8 \\ 14 \end{pmatrix} - \begin{pmatrix} -4 \\ 8 \\ 20 \end{pmatrix} \\
&= \begin{pmatrix} 200 + 4 \\ 800 - 8 \\ 1400 - 20 \end{pmatrix} \\
&= \begin{pmatrix} 204 \\ 792 \\ 1380 \end{pmatrix}.
\end{aligned}$$

□

Remark 3.19. The above exercise indicates that if we know how a matrix acts on a bunch of vectors, then we can determine how it acts on any vector that can be expressed as a *linear combination* of those vectors.

A **linear combination** of vectors $v_1, v_2, \dots, v_k$ is a vector of the form

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k,$$

where $c_1, c_2, \dots, c_k$ are real numbers⁵.

Note that any vector in $\mathbb{R}^2$ is a linear combination of the vectors

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

So, if A is a 3×2 matrix and we know how A acts on the vectors

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

that is, we know the products

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad A \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

then we can determine how A acts on any vector in $\mathbb{R}^2$. Indeed, for any $\begin{pmatrix} x \\ y \end{pmatrix}$ in $\mathbb{R}^2$, we have

$$\begin{pmatrix} x \\ y \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

and hence we obtain

$$\begin{aligned}
A \begin{pmatrix} x \\ y \end{pmatrix} &= A \left(x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \\
&= x A \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y A \begin{pmatrix} 0 \\ 1 \end{pmatrix}.
\end{aligned}$$

⁵One can also allow the c_i 's to be complex numbers, depending on the situation.

 Exercise

Show that every vector in $\mathbb{R}^2$ can be expressed as a linear combination of the vectors

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix}.$$

Solution. Note that for any $\begin{pmatrix} x \\ y \end{pmatrix}$ in $\mathbb{R}^2$, we have

$$\begin{pmatrix} x \\ y \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

that is, $\begin{pmatrix} x \\ y \end{pmatrix}$ is a linear combination of the vectors

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

So, it suffices⁶ to show that

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

can be expressed as linear combinations of the vectors

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix}.$$

Note that

$$7 \begin{pmatrix} 1 \\ 4 \end{pmatrix} - 4 \begin{pmatrix} 3 \\ 7 \end{pmatrix} = \begin{pmatrix} -5 \\ 0 \end{pmatrix},$$

$$3 \begin{pmatrix} 1 \\ 4 \end{pmatrix} - 1 \begin{pmatrix} 3 \\ 7 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \end{pmatrix},$$

which yields

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = -\frac{7}{5} \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \frac{4}{5} \begin{pmatrix} 3 \\ 7 \end{pmatrix},$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{3}{5} \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \frac{1}{5} \begin{pmatrix} 3 \\ 7 \end{pmatrix}.$$

□

 Remark

A careful reading of the above exercises and the remark indicates that if there are a few vectors $v_1, v_2, \dots, v_k$ in $\mathbb{R}^n$ such that every vector in $\mathbb{R}^n$ can be expressed as a linear combination of those vectors, then knowing how a matrix A acts on the vectors $v_1, v_2, \dots, v_k$ is sufficient to determine how A acts on any vector in $\mathbb{R}^n$.

Exercise 3.20. Note that

⁶How does it suffice?

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix},$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix},$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix}$$

hold. The vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

are called the **standard basis vectors** of $\mathbb{R}^3$. The standard basis vectors of $\mathbb{R}^2$ are

$$e_1 := \begin{pmatrix} 1 \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

In general, the vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

are called the **standard basis vectors** of $\mathbb{R}^n$.



Example

Note that

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix},$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

hold. The vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

are called the **standard basis vectors** of $\mathbb{R}^2$.

Exercise 3.21. If A is an $m \times n$ matrix, then show that for all $i = 1, 2, \dots, n$, the product Ae_i is equal to the i -th column of A , where e_i denotes the i -th standard basis vector of $\mathbb{R}^n$, that is,

$$e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

with 1 in the i -th position, and 0's elsewhere.

i Remark

[Exercise 3.21](#) shows that the columns of a matrix A can be obtained by multiplying A with the standard basis vectors. Further, if A is an $m \times n$ matrix, and we know how A acts on the standard basis vectors of $\mathbb{R}^n$, then we can determine all the entries of A , and hence we can determine how A acts on any vector v in $\mathbb{R}^n$. Note that this is achieved by determining the entries of A .

Moreover, the first remark in [Section 3.2.4](#) indicates that if we know how A acts on the standard basis vectors of $\mathbb{R}^n$, then using that every element of $\mathbb{R}^n$ can be expressed as a linear combination of the standard basis vectors, we can determine how A acts on any vector v in $\mathbb{R}^n$.

Furthermore, if $v_1, v_2, \dots, v_k$ are elements of $\mathbb{R}^n$, then knowing how a matrix A acts on $v_1, v_2, \dots, v_k$ is sufficient to determine how A acts on any vector which can be expressed as a linear combination of $v_1, \dots, v_k$.

Let us revisit [Exercise 3.16](#), under the light of [Exercise 3.21](#).

Exercise 3.22. Compute B^2 where B is the following $n \times n$ matrix

$$B = \begin{pmatrix} 0 & \cdots & 0 & 1 \\ 0 & \cdots & 1 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 1 & \cdots & 0 & 0 \end{pmatrix}.$$

Solution. Let $e_1, e_2, \dots, e_n$ denote the standard basis vectors of $\mathbb{R}^n$. Note that B sends $e_1, e_2, \dots, e_{n-1}, e_n$ to the vectors $e_n, e_{n-1}, \dots, e_2, e_1$ respectively. That is,

$$Be_i = e_{n+1-i}$$

holds for all $1 \leq i \leq n$. Hence, for any integer $1 \leq i \leq n$, it follows that

$$\begin{aligned} B^2 e_i &= B(Be_i) \\ &= B(e_{n+1-i}) \\ &= e_{n+1-(n+1-i)} \\ &= e_i. \end{aligned}$$

This shows that⁷

$$B^2 = I_n.$$

□

⁷How does it follow? Does [Exercise 3.21](#) help?

§3.3 System of linear equations



Exercise

Solve the system of equations

$$\begin{aligned} 2x + 3y &= 8, \\ 5x + 7y &= 19. \end{aligned}$$

Solution. Multiplying the first equation by 7, and the second equation by 3, and taking the difference of the equations obtained, we get

$$7 \cdot 2x - 3 \cdot 5x = 7 \cdot 8 - 3 \cdot 19,$$

which yields

$$x = \frac{7 \cdot 8 - 3 \cdot 19}{7 \cdot 2 - 3 \cdot 5}.$$

Similarly, multiplying the first equation by 5, and the second equation by 2, and taking the difference of the equations obtained, we get

$$5 \cdot 3y - 2 \cdot 7y = 5 \cdot 8 - 2 \cdot 19,$$

which yields

$$y = \frac{5 \cdot 8 - 2 \cdot 19}{5 \cdot 3 - 2 \cdot 7}.$$

□

Alternate solution. Note that the given system of equations can be rewritten as

$$\begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8 \\ 19 \end{pmatrix}, \quad (2)$$

which can be thought as a concise way of putting the information

$$\begin{pmatrix} 2x + 3y \\ 5x + 7y \end{pmatrix} = \begin{pmatrix} 8 \\ 19 \end{pmatrix}. \quad (3)$$

Recall that in the previous solution,

- to solve for x , the first equation was multiplied by 7, and the second equation by 3, and next, we considered the difference of the equations obtained,
- to solve for y , the first equation was multiplied by 5, and the second equation was multiplied by 2, and then, we considered the difference of the equations obtained.

The two steps can be put together in the matrix form as

$$\begin{pmatrix} 7 & -3 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} 2x + 3y \\ 5x + 7y \end{pmatrix}.$$

Using Equation 3, it yields

$$\begin{pmatrix} 7 & -3 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} 2x + 3y \\ 5x + 7y \end{pmatrix} = \begin{pmatrix} 7 & -3 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} 8 \\ 19 \end{pmatrix}.$$

The advantage of this process is that, first y gets eliminated in the first step and helps to find x , and in the next step, x gets eliminated and leads to finding y . Stated in matrix form, we have that

$$\begin{pmatrix} 7 & -3 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} 2x + 3y \\ 5x + 7y \end{pmatrix} = \begin{pmatrix} (7 \cdot 2 - 3 \cdot 5)x \\ (5 \cdot 3 - 2 \cdot 7)y \end{pmatrix}.$$

We could have also written

$$\begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 2x + 3y \\ 5x + 7y \end{pmatrix} = \begin{pmatrix} (7 \cdot 2 - 3 \cdot 5)x \\ (-5 \cdot 3 + 2 \cdot 7)y \end{pmatrix}.$$

Thus, rewriting the given system of equations in the matrix form as in Equation 3, and multiplying it from the left by the matrix

$$\begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix},$$

we obtain

$$\begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 2x + 3y \\ 5x + 7y \end{pmatrix} = \begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 8 \\ 19 \end{pmatrix},$$

which gives

$$\begin{pmatrix} (7 \cdot 2 - 3 \cdot 5)x \\ (-5 \cdot 3 + 2 \cdot 7)y \end{pmatrix} = \begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 8 \\ 19 \end{pmatrix},$$

that is, we have

$$(7 \cdot 2 - 3 \cdot 5) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 8 \\ 19 \end{pmatrix},$$

and hence, the solution to the given system of equations is given by

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{7 \cdot 2 - 3 \cdot 5} \begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 8 \\ 19 \end{pmatrix}.$$

□

Summarizing the above. The given system of equations can be rewritten as

$$\begin{pmatrix} 2x + 3y \\ 5x + 7y \end{pmatrix} = \begin{pmatrix} 8 \\ 19 \end{pmatrix}, \quad (4)$$

or equivalently,

$$\begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8 \\ 19 \end{pmatrix}, \quad (5)$$

Multiplying the above from the left by the matrix

$$\frac{1}{7 \cdot 2 - 3 \cdot 5} \begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix}, \quad (6)$$

we obtain

$$\frac{1}{7 \cdot 2 - 3 \cdot 5} \begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 2x + 3y \\ 5x + 7y \end{pmatrix} = \frac{1}{7 \cdot 2 - 3 \cdot 5} \begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 8 \\ 19 \end{pmatrix},$$

which shows that the solution to the given system of equations is

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{7 \cdot 2 - 3 \cdot 5} \begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 8 \\ 19 \end{pmatrix}.$$

□

i Remark

In the above, Equation 5 was multiplied by the matrix as in Equation 6, since multiplying the matrix

$$\begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$$

as in Equation 5 from the left by the matrix

$$\frac{1}{7 \cdot 2 - 3 \cdot 5} \begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix},$$

as in Equation 6 yields

$$\begin{aligned} \frac{1}{7 \cdot 2 - 3 \cdot 5} \begin{pmatrix} 7 & -3 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix} &= \frac{1}{7 \cdot 2 - 3 \cdot 5} \begin{pmatrix} 7 \cdot 2 - 3 \cdot 5 & 0 \\ 0 & -5 \cdot 3 + 2 \cdot 7 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \end{aligned}$$

which is the 2×2 identity matrix.

§3.4 Invertible matrices

Definition

Let A be an $n \times n$ matrix. If there is an $n \times n$ matrix B such that

$$AB = BA = I_n,$$

then B is called an **inverse** of A . If A admits an inverse, then A is called **invertible**.

Exercise 3.23. Show that if B, C are inverses of an $n \times n$ matrix A , then $B = C$.

Solution. Using $AB = I_n$, we obtain

$$CAB = CI_n = C.$$

Using $CA = I_n$, we get

$$CAB = I_n B = B.$$

This shows that $B = C$. □

i Remark

If A is an $n \times n$ matrix, and A admits an inverse, then it is called **the** inverse of A , and is denoted by A^{-1} .

Exercise 3.24. Let A, B be invertible $n \times n$ matrices. Show that AB is invertible, and that

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Compare the above exercise with Exercise 3.8.

Solution. Note that

$$\begin{aligned}
(AB)(B^{-1}A^{-1}) &= A(BB^{-1})A^{-1} \\
&= AI_n A^{-1} \\
&= AA^{-1} \\
&= I_n,
\end{aligned}$$

and

$$\begin{aligned}
(B^{-1}A^{-1})(AB) &= B^{-1}(A^{-1}A)B \\
&= B^{-1}I_n B \\
&= B^{-1}B \\
&= I_n.
\end{aligned}$$

This shows that AB is invertible, and that

$$(AB)^{-1} = B^{-1}A^{-1}.$$

□

Lemma 3.25. *Let*

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

be a 2×2 matrix. Show that the following statements are equivalent.

- *A is invertible,*
- *$ad - bc$ is nonzero.*

Moreover, if A is invertible, then its inverse is equal to

$$\frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Proof. Note that

$$\begin{aligned}
\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} &= \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix}, \\
\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} &= \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix}
\end{aligned}$$

hold. Assume that if A is invertible. It follows that

$$\begin{aligned}
(ad - bc)A^{-1} &= \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix} A^{-1} \\
&= \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} A^{-1} \\
&= \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} AA^{-1} \\
&= \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} I_2 \\
&= \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.
\end{aligned}$$

Note that if $ad - bc = 0$, then a, b, c, d are equal to 0, which implies $A = 0$, which is impossible since $AA^{-1} = I_2$. Hence, $ad - bc$ is nonzero, and

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Conversely, note that if $ad - bc \neq 0$, then it follows that the matrix

$$\frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

is the inverse⁸ of A . □



Definition

If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is a 2×2 matrix, then its **determinant** is denoted by $\det A$, and is defined as $ad - bc$.



Remark

The determinant of a 2×2 matrix can be used to determine whether the matrix is invertible or not. Indeed, by [Lemma 3.25](#), the following statements are equivalent.

- The matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is invertible.
- The determinant $\det\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right)$ is nonzero.

Moreover, if $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is invertible, then its inverse is given by

$$\frac{1}{\det\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Exercise 3.26. If A, B are 2×2 matrices, then show that

$$\det(AB) = \det(A) \det(B).$$

Solution. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

and

$$B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$$

be 2×2 matrices. Note that

$$AB = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}.$$

Using the definition of determinant, we obtain

$$\begin{aligned} \det(AB) &= (ae)(cf) - (af)(ce) + (bg)(dh) - (bh)(dg) \\ &= acef - acef + bdgh - bdgh \\ &= (ad - bc)(eh - fg) \\ &= \det(A) \det(B). \end{aligned}$$
□

Exercise 3.27. Solve the system of equations

$$\begin{aligned} 5x + 2y &= 11, \\ 3x + 4y &= 8. \end{aligned}$$

⁸Why?

Solution. Note that the given system of equations can be rewritten as

$$\begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 11 \\ 8 \end{pmatrix}, \quad (7)$$

Note that

$$\det \begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix} = 5 \cdot 4 - 2 \cdot 3 = 14,$$

which is nonzero. Hence, the matrix $\begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix}$ is invertible, and

$$\begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix}^{-1} = \frac{1}{14} \begin{pmatrix} 4 & -2 \\ -3 & 5 \end{pmatrix}.$$

Multiplying Equation 7 from the left by the inverse of the matrix $\begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix}$, we obtain

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 4 & -2 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} 11 \\ 8 \end{pmatrix},$$

which yields

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 44 - 16 \\ -33 + 40 \end{pmatrix} = \begin{pmatrix} 2 \\ \frac{1}{2} \end{pmatrix}.$$

□

Exercise 3.28. Solve the system of equations

$$\begin{aligned} x + 2y + 3z &= 1, \\ 4x + 5y + 6z &= 2, \\ 7x + 8y + 10z &= 3. \end{aligned}$$

Exercise 3.29. Let A, B be 2×2 matrices. Show that if AB is invertible, then both A and B are invertible.

Exercise 3.30. Let A be an $n \times n$ matrix. If P is an invertible $n \times n$ matrix, then show that

$$(PAP^{-1})^k = PA^kP^{-1}$$

holds for any positive integer k .

Solution. We prove the statement by induction. Let P be an invertible $n \times n$ matrix. For a positive integer k , let $P(k)$ denote the statement

$$(PAP^{-1})^k = PA^kP^{-1}.$$

Note that $P(1)$ holds. Assume that $P(k)$ holds for some positive integer k . Then, using the induction hypothesis, we obtain

$$\begin{aligned} (PAP^{-1})^{k+1} &= (PAP^{-1})^k (PAP^{-1}) \\ &= (PA^kP^{-1})(PAP^{-1}) \\ &= PA^k(P^{-1}P)AP^{-1} \\ &= PA^kI_nAP^{-1} \\ &= PA^{k+1}P^{-1}. \end{aligned}$$

This shows that $P(k+1)$ holds. Hence, by the principle of mathematical induction, $P(k)$ holds for all positive integers k . □

§3.5 Systems of linear equations again

§3.5.1 Linear combination of the columns of a matrix

Here are further examples of systems of linear equations.

System of linear equations	Matrix form	Solution(s)	Associated matrix
$x + 2y = 1,$ $3x - 5y = -7$	$\begin{pmatrix} 1 & 2 \\ 3 & -5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -7 \end{pmatrix}$	Unique	Invertible
$x + 2y = 1,$ $2x + 4y = 2$	$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$	Infinitely many ⁹	Not invertible
$x + 2y = 1,$ $3x + 6y = 2$	$\begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$	None	Not invertible

? Question

Which of such systems of equations can be solved?

Remark 3.31. Consider the following system of linear equations.

$$12345x + 67890y = 11111,$$

$$54321x + 9876y = 22222.$$

To solve it, we need to find a vector $\begin{pmatrix} x \\ y \end{pmatrix}$ in $\mathbb{R}^2$ satisfying

$$\begin{pmatrix} 12345 & 67890 \\ 54321 & 9876 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 11111 \\ 22222 \end{pmatrix}.$$

In other words, we need to find real numbers x, y such that

$$\begin{aligned} \begin{pmatrix} 12345 & 67890 \\ 54321 & 9876 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 12345 & 67890 \\ 54321 & 9876 \end{pmatrix} (xe_1 + ye_2) \\ &= x \begin{pmatrix} 12345 & 67890 \\ 54321 & 9876 \end{pmatrix} e_1 + y \begin{pmatrix} 12345 & 67890 \\ 54321 & 9876 \end{pmatrix} e_2 \\ &= x \begin{pmatrix} 12345 \\ 54321 \end{pmatrix} + y \begin{pmatrix} 67890 \\ 9876 \end{pmatrix} \end{aligned}$$

is equal to $\begin{pmatrix} 11111 \\ 22222 \end{pmatrix}$, where e_1, e_2 are the standard basis vectors of $\mathbb{R}^2$.

It is left to determining whether some linear combination of the columns of the matrix $\begin{pmatrix} 12345 & 67890 \\ 54321 & 9876 \end{pmatrix}$ is equal to $\begin{pmatrix} 11111 \\ 22222 \end{pmatrix}$. If such a linear combination exists, then the system of equations has a solution, and if such a linear combination does not exist, then the system of equations does not have a solution. If there is more than one such linear combination, then the system of equations has more than one solution, etc.

⁹Why?

i Remark

The above discussion shows that solving a system of linear equations is equivalent to determining whether **some linear combination** of the columns of a matrix is equal to the given vector. This is the reason why the notions¹⁰ of *column space* and *linear span of vectors* are important in the study of systems of linear equations.

i Remark

Consider a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, and a vector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ where $a, b, c, d, \alpha, \beta$ are real numbers. If $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is **not** a linear combination of the columns of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, that is, if $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is **not** a linear combination of the vectors $\begin{pmatrix} a \\ c \end{pmatrix}$ and $\begin{pmatrix} b \\ d \end{pmatrix}$, then the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

admits **no** solution.

Lemma 3.32. Let A be a 2×2 matrix, and let $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ be a vector of $\mathbb{R}^2$. Show that the following statements are equivalent.

- The vector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is a linear combination of the columns of the matrix A .
- The system of equations

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

admits at least one solution.

Proof. Assume that $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is a linear combination of the columns of the matrix A . This shows that there are real numbers x_0, y_0 such that

$$A \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Hence, the system of equations

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

admits at least one solution.

Conversely, assume that the system of equations

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

admits at least one solution, say $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$. It follows that

$$A \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

This shows that $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is a linear combination of the columns of the matrix A . □

§3.5.2 Homogeneous system of linear equations

Let us first consider the solutions to the system of equations

¹⁰The notions of column space and linear span of vectors will be introduced formally at a later stage.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (8)$$

Note that $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is a linear combination of the columns of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, since

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0 \begin{pmatrix} a \\ c \end{pmatrix} + 0 \begin{pmatrix} b \\ d \end{pmatrix}.$$

This shows that $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is a solution to Equation 8.

Case 1

Let us first consider the case that the matrix A is invertible.

Note that under this hypothesis, Equation 8 has no solution other than $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$. Indeed, if $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ is another solution to Equation 8, then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This shows that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which yields

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Hence, if A is invertible, then the only solution to Equation 8 is $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Observation

If Equation 8 has at least one solution other than $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, then Equation 8 has infinitely many solutions. Indeed, if $\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ is a solution to Equation 8 other than $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, then for any real number t , the vector $t \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ is also a solution to Equation 8, and hence, Equation 8 has infinitely many solutions.

Case 2

Now let us consider the case that the matrix A is **not** invertible. By Lemma 3.25, we have $ad - bc = 0$, this implies that

$$\begin{aligned} d \begin{pmatrix} a \\ c \end{pmatrix} - c \begin{pmatrix} b \\ d \end{pmatrix} &= \begin{pmatrix} ad - bc \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \\ b \begin{pmatrix} a \\ c \end{pmatrix} - a \begin{pmatrix} b \\ d \end{pmatrix} &= \begin{pmatrix} 0 \\ bc - ad \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \end{aligned}$$

and consequently,

$$\begin{aligned} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d \\ -c \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \\ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} b \\ -a \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \end{aligned}$$

This shows that if A is **not** invertible, then the vectors $\begin{pmatrix} d \\ -c \end{pmatrix}$ and $\begin{pmatrix} b \\ -a \end{pmatrix}$ are solutions to Equation 8.

Subcase 2a

If the vectors $\begin{pmatrix} d \\ -c \end{pmatrix}$ and $\begin{pmatrix} b \\ -a \end{pmatrix}$ are equal to the zero vector, equivalently, if a, b, c, d are all equal to 0, then any vector $\begin{pmatrix} x \\ y \end{pmatrix}$ of $\mathbb{R}^2$ is a solution to Equation 8.

Subcase 2b

If not both of the vectors $\begin{pmatrix} d \\ -c \end{pmatrix}$ and $\begin{pmatrix} b \\ -a \end{pmatrix}$ are equal to the zero vector (equivalently, at least one of them is nonzero) and A is not invertible, then Equation 8 admits a solution other than $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, and moreover, not all vectors of $\mathbb{R}^2$ are solutions to Equation 8 since not all of $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are solutions to Equation 8.

Lemma 3.33. Consider the system of linear equations

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad (9)$$

where A is a 2×2 matrix. The following statements are equivalent.

- The matrix A is invertible.
- The system of equations as in Equation 9 admits a unique solution, that is, it admits no solution other than the trivial solution $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.
- The trivial linear combination of the columns of the matrix A is the only linear combination of the columns of A that is equal to $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Furthermore, the following statements are equivalent.

- Any element of $\mathbb{R}^2$ is a solution to Equation 9.
- The matrix A is the zero matrix.

Moreover, the following statements are equivalent too.

- The system of equations as in Equation 9 admits a solution other than the trivial solution $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, and not all elements of $\mathbb{R}^2$ are solutions to Equation 9.
- The matrix A is not invertible and nonzero.

Proof. In the following, the first and the second column of A are denoted by C_1 and C_2 respectively. Write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Let us first show that the first three statements are equivalent. Assume that A is invertible. If $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ is a solution to Equation 9, then

$$A \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This shows that

$$A^{-1}A \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which yields

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This implies that the solution $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is the only solution to Equation 9. This shows that the first statement implies the second statement. Assume that the second statement holds, that is, Equation 9 admits no solution other than $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$. If s, t are real numbers such that

$$sC_1 + tC_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

then

$$A \begin{pmatrix} s \\ t \end{pmatrix} = sC_1 + tC_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This shows that $\begin{pmatrix} s \\ t \end{pmatrix}$ is a solution to Equation 9, and hence, $s = t = 0$. This shows that the second statement implies the third statement. Assume that the third statement holds, that is, the trivial linear combination of the columns of A is the only linear combination of the columns of A that is equal to $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$. If A is not invertible, then by Lemma 3.25, we have $ad - bc = 0$, which implies that

$$\begin{aligned} dC_1 - cC_2 &= \begin{pmatrix} ad - bc \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \\ -bC_1 + aC_2 &= \begin{pmatrix} 0 \\ ad - bc \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \end{aligned}$$

and consequently, a, b, c, d are equal to 0, and hence, any linear combination of the columns of A is equal to $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, contradicting the third statement. This shows that if the third statement holds, then A is invertible, and hence, the third statement implies the first statement. This proves the equivalence of the first three statements.

For the equivalence of the next two statements, note that the following statements are equivalent.

- any element of $\mathbb{R}^2$ is a solution to Equation 9
- $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are solutions to Equation 9
- the matrix A is the zero matrix.

□

Exercise 3.34. Let A be a 2×2 matrix. Let V be the set of solutions to the system of equations

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

that is,

$$V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}.$$

Show that for any $u, v \in V$ and $c \in \mathbb{R}$, the element $u + v$ of $\mathbb{R}^2$ lies in V , and the element cu also lies in V .

Solution. Let u, v be elements of V , and let c be a real number. Note that

$$A(u + v) = Au + Av = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

and

$$A(cu) = c(Au) = c \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This shows that both $u + v$ and cu lie in V .

□

i Remark

What about the difference of two elements of V ? If u, v are elements of V and c, d are real numbers, then what can be said about the element $cu + dv$ of $\mathbb{R}^2$?

Exercise 3.35. Let A be a 2×2 matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A , that is¹¹,

$$V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} s \\ t \end{pmatrix} \text{ for some } s, t \in \mathbb{R} \right\}.$$

Show that for any $u, v \in V$ and $c \in \mathbb{R}$, the element $u + v$ of $\mathbb{R}^2$ lies in V , and the element cu also lies in V .

Solution. Let u, v be elements of V , and let c be a real number. By the definition of V , there exist real numbers s_1, t_1, s_2, t_2 such that

$$u = A \begin{pmatrix} s_1 \\ t_1 \end{pmatrix},$$

$$v = A \begin{pmatrix} s_2 \\ t_2 \end{pmatrix}.$$

Note that

$$u + v = A \begin{pmatrix} s_1 \\ t_1 \end{pmatrix} + A \begin{pmatrix} s_2 \\ t_2 \end{pmatrix} = A \left(\begin{pmatrix} s_1 \\ t_1 \end{pmatrix} + \begin{pmatrix} s_2 \\ t_2 \end{pmatrix} \right) = A \begin{pmatrix} s_1 + s_2 \\ t_1 + t_2 \end{pmatrix},$$

and

$$cu = c \left(A \begin{pmatrix} s_1 \\ t_1 \end{pmatrix} \right) = A \left(c \begin{pmatrix} s_1 \\ t_1 \end{pmatrix} \right) = A \begin{pmatrix} cs_1 \\ ct_1 \end{pmatrix}.$$

This shows that both $u + v$ and cu lie in V . □

Exercise 3.36. Let A be an $m \times n$ matrix. Let V be the set of solutions to the system of equations $Au = 0$,

that is,

$$V = \{v \in \mathbb{R}^n : Av = 0\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $\mathbb{R}^m$ lies in V .

Solution. Let u, v be elements of V , and let c, d be real numbers. Note that

$$A(cu + dv) = c(Au) + d(Av) = c \cdot 0 + d \cdot 0 = 0.$$

This shows that $cu + dv$ lies in V . □

Exercise 3.37. Let A be an $m \times n$ matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A , that is,

$$V = \{v \in \mathbb{R}^m : v = Au \text{ for some } u \in \mathbb{R}^n\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $\mathbb{R}^m$ lies in V .

¹¹Observe that

$$A \begin{pmatrix} s \\ t \end{pmatrix} = sC_1 + tC_2,$$

where C_1, C_2 denote the first and second columns of A respectively.

Solution. Let u, v be elements of V , and let c, d be real numbers. By the definition of V , there exist vectors $u_1, u_2 \in \mathbb{R}^n$ such that

$$\begin{aligned} u &= Au_1, \\ v &= Au_2. \end{aligned}$$

Note that

$$cu + dv = c(Au_1) + d(Au_2) = A(cu_1) + A(du_2) = A(cu_1 + du_2).$$

This shows that $cu + dv$ lies in V . □

§3.5.3 General system of linear equations

If $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is a linear combination of the columns of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then there exist real numbers x_0, y_0 such that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

This shows that the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

has at least one solution, namely $\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$. If $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ is another solution to the above system of equations, then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

This shows that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}.$$

Hence,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} - \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} \right) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

holds. This shows that the difference of any two solutions to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

is a solution to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Conversely, if $\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ is a solution to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

then

$$\begin{aligned} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \left(\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \right) &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \\ &= \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \end{aligned}$$

This shows that if $\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ is a solution to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

then $\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ is also a solution to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

In summary, if $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is a linear combination of the columns of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then the solutions to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

are in one-to-one correspondence with the solutions to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Remark

The above discussion shows that if $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is a linear combination of the columns of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then the solutions to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

are in one-to-one correspondence with the solutions to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Hence, to determine the number of solutions to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

it suffices to determine the number of solutions to the system of equations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Lemma 3.38. Let A be a 2×2 matrix, and let $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ be a vector of $\mathbb{R}^2$. Assume that $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is a linear combination of the columns of the matrix A . Show that the solutions to the system of equations

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

are in one-to-one correspondence with the solutions to the system of equations

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

More specifically, for any fixed solution $\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$ to the system of equations

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix},$$

the map

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

is a bijection from the set of solutions to the system of equations

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

to the set of solutions to the system of equations

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

having the following map as its inverse:

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}.$$

Proof. Deferred. □

Remark

The terms “one-to-one correspondence” and “bijection” will be introduced formally at a later stage. The proof of the above lemma is deferred until then.

§3.6 Special types of matrices

Recall that a matrix is called a **square** matrix if it has the same number of rows and columns.

Definition

A square matrix A is called **diagonal** if all its non-diagonal entries are equal to zero.

Example

The following matrices are diagonal.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \begin{pmatrix} 5 & 0 \\ 0 & -7 \end{pmatrix}, \begin{pmatrix} i & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \pi \end{pmatrix}.$$

Some non-diagonal matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Definition

A square matrix A is called **scalar** if it is a diagonal matrix and all its diagonal entries are equal. In other words, an $n \times n$ matrix A is called **scalar** if

$$A = \lambda I_n$$

for some scalar λ .

**Example**

The following matrices are scalar.

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Some non-scalar matrices are as follows.

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 5 & 0 & 0 \\ 0 & -7 & 0 \\ 0 & 0 & \pi \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

**Definition**

A square matrix A is called **triangular** if all its entries below the main diagonal are equal to zero, or if all its entries above the main diagonal are equal to zero.

A triangular matrix A is called **upper-triangular** if all its entries below the main diagonal are equal to zero.

A triangular matrix A is called **lower-triangular** if all its entries above the main diagonal are equal to zero.

**Example**

The following matrices are upper-triangular.

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 8 \\ 0 & 9 \end{pmatrix}.$$

The following matrices are lower-triangular.

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 0 \\ 8 & 9 \end{pmatrix}.$$

Some non-triangular matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 5 & 0 & 6 \end{pmatrix}.$$

**Definition**

A square matrix A is called **symmetric** if

$$A^T = A.$$

**Example**

The following matrices are symmetric.

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 8 \\ 8 & 9 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Some non-symmetric matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

**Definition**

A square matrix A is called **skew-symmetric** if

$$A^T = -A.$$

**Example**

The following matrices are skew-symmetric.

$$\begin{pmatrix} 0 & 2 & 3 \\ -2 & 0 & 5 \\ -3 & -5 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 8 \\ -8 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Here are some matrices that are not skew-symmetric.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 3.39. Let A be a square matrix. Show that if A is symmetric and skew-symmetric, then A is the zero matrix.

Solution. Assume that A is symmetric and skew-symmetric. It follows that

$$A^T = A,$$

$$A^T = -A.$$

This shows that

$$A = -A,$$

which yields

$$2A = 0,$$

and hence, A is the zero matrix. □

Exercise 3.40. Let A be a square matrix. Show that A can be uniquely expressed as the sum of a symmetric matrix and a skew-symmetric matrix.

Solution. Note that

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T).$$

Note that the matrix $\frac{1}{2}(A + A^T)$ is symmetric, and that the matrix $\frac{1}{2}(A - A^T)$ is skew-symmetric. This shows that A can be expressed as the sum of a symmetric matrix and a skew-symmetric matrix.

Assume that

$$A = S_1 + K_1 = S_2 + K_2,$$

where S_1, S_2 are symmetric matrices, and K_1, K_2 are skew-symmetric matrices. It follows that

$$S_1 - S_2 = K_2 - K_1.$$

Note that the left-hand side of the above equality is a symmetric matrix, and that the right-hand side of the above equality is a skew-symmetric matrix. This shows that each of $S_1 - S_2, K_2 - K_1$ is both symmetric and skew-symmetric. Since the only matrix that is both symmetric and skew-symmetric is the zero matrix, we have $S_1 - S_2 = K_2 - K_1 = 0$. This yields

$$S_1 = S_2,$$

$$K_1 = K_2.$$

This implies that any square matrix A can be **uniquely** expressed as the sum of a symmetric matrix and a skew-symmetric matrix. $\square$



Definition

An $n \times n$ matrix A with **real entries** is called **orthogonal** if

$$AA^T = A^T A = I_n.$$



Example

The following matrices are orthogonal.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Some non-orthogonal matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

 Remark

There are infinitely many orthogonal matrices. For instance, for any real number θ , the matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

is an orthogonal matrix since

$$\begin{aligned} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}^T \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} &= \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \\ &= \begin{pmatrix} \cos^2 \theta + \sin^2 \theta & 0 \\ 0 & \cos^2 \theta + \sin^2 \theta \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \end{aligned}$$

and

$$\begin{aligned} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}^T &= \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \\ &= \begin{pmatrix} \cos^2 \theta + \sin^2 \theta & 0 \\ 0 & \cos^2 \theta + \sin^2 \theta \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

Exercise 3.41. Show that for any $n \geq 2$, there are infinitely many $n \times n$ orthogonal matrices.

Exercise 3.42. Let A be an orthogonal matrix. Show that A is invertible and that

$$A^{-1} = A^T.$$

Solution. Let A be an $n \times n$ orthogonal matrix. It follows that

$$AA^T = A^T A = I_n,$$

which shows that A is invertible and that

$$A^{-1} = A^T.$$

□

Exercise 3.43. Let A, B be orthogonal matrices of the same size. Show that the matrix product AB is also an orthogonal matrix.

Solution. Let A, B be $n \times n$ orthogonal matrices. Note that

$$(AB)(AB)^T = AB B^T A^T = A I_n A^T = A A^T = I_n,$$

and similarly,

$$(AB)^T(AB) = B^T A^T AB = B^T I_n B = B^T B = I_n.$$

This shows that the matrix product AB is also an orthogonal matrix.

□

 Definition

If A is a matrix with complex entries, then the **complex conjugate** of A , denoted by $\overline{A}$, is defined as the matrix of the same size, obtained by taking the complex conjugate of each entry of A .

**Example**

The complex conjugate of the matrix

$$\begin{pmatrix} 1 + 2i & 3 - 4i \\ -i & 5 \end{pmatrix}$$

is the matrix

$$\begin{pmatrix} 1 - 2i & 3 + 4i \\ i & 5 \end{pmatrix}.$$

Exercise 3.44. Let A, B be square matrices with complex entries. Show that

$$\overline{AB} = \overline{A} \overline{B}.$$

Exercise 3.45. Let A be a square matrix with complex entries. Show that

$$\overline{A^T} = (\overline{A})^T.$$

**Definition**

If A is a matrix with complex entries, then the **conjugate transpose** of A , denoted by A^* , is defined as the transpose of the matrix obtained by taking the complex conjugate of each entry of A , that A^* is equal to $\overline{A^T}$.

**Example**

The conjugate transpose of the matrix

$$\begin{pmatrix} 1 + 2i & 3 - 4i \\ -i & 5 \end{pmatrix}$$

is the matrix

$$\begin{pmatrix} 1 - 2i & i \\ 3 + 4i & 5 \end{pmatrix}.$$

Compare the following exercises with [Exercise 3.6](#), [Exercise 3.7](#), [Exercise 3.8](#).

Exercise 3.46. Let A, B be square matrices of the same size with complex entries. Show that

$$(A + B)^* = A^* + B^*.$$

Solution. Note that

$$(A + B)^* = (\overline{A + B})^T = (\overline{A} + \overline{B})^T = \overline{A}^T + \overline{B}^T = A^* + B^*.$$

□

Exercise 3.47. Let A be a matrix with complex entries. Show that

$$(A^*)^* = A.$$

Solution. Note that

$$(A^*)^* = (\overline{A^T})^* = (\overline{A^T})^* = \overline{\overline{A^T}}^T = (A^T)^T = A.$$

□

Exercise 3.48. Let A, B be matrices with complex entries. Show that

$$(AB)^* = B^* A^*.$$

Solution. Note that

$$(AB)^* = (\overline{AB})^T = (\overline{A} \overline{B})^T = \overline{B}^T \overline{A}^T = B^* A^*.$$

□



Definition

A square matrix A with complex entries is called **hermitian** if

$$A^* = A,$$

where A^* denotes the conjugate transpose of A .



Example

The following matrices are hermitian.

$$\begin{pmatrix} 1 & 2+i \\ 2-i & 3 \end{pmatrix}, \begin{pmatrix} 5 & i \\ -i & 7 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Some non-hermitian matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 3.49. Let A be an $n \times n$ hermitian matrix. Show that all the diagonal entries of A are real numbers. More generally, show that for any $1 \leq i, j \leq n$, the (i, j) -entry and the (j, i) -entry of A are complex conjugates of each other.

Exercise 3.50. Let A, B be hermitian matrices of the same size. Show that the matrix $A + B$ is also a hermitian matrix.



Definition

A square matrix A with complex entries is called **skew-hermitian** if

$$A^* = -A,$$

where A^* denotes the conjugate transpose of A .



Example

The following matrices are skew-hermitian.

$$\begin{pmatrix} 0 & 2+i \\ -2+i & 0 \end{pmatrix}, \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Some non-skew-hermitian matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 3.51. State and prove an analogue of [Exercise 3.49](#) for skew-hermitian matrices.

Exercise 3.52. Let A, B be skew-hermitian matrices of the same size. Show that the matrix $A + B$ is also a skew-hermitian matrix.

Exercise 3.53. Let A be a square matrix with complex entries. Show that if A is hermitian and skew-hermitian, then A is the zero matrix.

Solution. Assume that A is hermitian and skew-hermitian. It follows that

$$\begin{aligned} A^* &= A, \\ A^* &= -A. \end{aligned}$$

This shows that

$$A = -A,$$

which yields

$$2A = 0,$$

and hence, A is the zero matrix. $\square$

Exercise 3.54. Let A be a square matrix with complex entries. Show that A can be uniquely expressed as the sum of a hermitian matrix and a skew-hermitian matrix.

Solution. Note that

$$A = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*).$$

Note that the matrix $\frac{1}{2}(A + A^*)$ is hermitian, and that the matrix $\frac{1}{2}(A - A^*)$ is skew-hermitian. This shows that A can be expressed as the sum of a hermitian matrix and a skew-hermitian matrix.

Assume that

$$A = H_1 + K_1 = H_2 + K_2,$$

where H_1, H_2 are hermitian matrices, and K_1, K_2 are skew-hermitian matrices. It follows that

$$H_1 - H_2 = K_2 - K_1.$$

Note that the left-hand side of the above equality is a hermitian matrix, and that the right-hand side of the above equality is a skew-hermitian matrix. This shows that each of $H_1 - H_2, K_2 - K_1$ is both hermitian and skew-hermitian. Since the only matrix that is both hermitian and skew-hermitian is the zero matrix, we have $H_1 - H_2 = K_2 - K_1 = 0$. This yields

$$\begin{aligned} H_1 &= H_2, \\ K_1 &= K_2. \end{aligned}$$

This implies that any square matrix A with complex entries can be **uniquely** expressed as the sum of a hermitian matrix and a skew-hermitian matrix. $\square$



Definition

An $n \times n$ matrix A with complex entries is called **unitary** if

$$AA^* = A^*A = I_n.$$

**Example**

The following matrices are unitary.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Some non-unitary matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 3.55. Let A be a unitary matrix. Show that A is invertible and that

$$A^{-1} = A^*.$$

Solution. Let A be a unitary matrix. It follows that

$$AA^* = A^*A = I_n,$$

which shows that A is invertible and that

$$A^{-1} = A^*.$$

□

Exercise 3.56. Let A, B be matrices of the same size with complex entries. Show that

$$(AB)^* = B^*A^*.$$

Solution. Let A, B be matrices of the same size with complex entries. Note that

$$(AB)^* = \overline{(AB)}^T = \overline{B^T A^T} = \overline{B^T} \overline{A^T} = B^* A^*.$$

□

Exercise 3.57. Let A, B be unitary matrices of the same size. Show that the matrix product AB is also a unitary matrix.

Solution. Let A, B be $n \times n$ unitary matrices. Note that

$$(AB)(AB)^* = ABB^*A^* = AI_nA^* = AA^* = I_n,$$

and similarly,

$$(AB)^*(AB) = B^*A^*AB = B^*I_nB = B^*B = I_n.$$

This shows that the matrix product AB is also a unitary matrix.

□

Exercise 3.58. Show that any orthogonal matrix is a unitary matrix. Is the converse true?

Solution. Let A be an orthogonal matrix. It follows that

$$A^T A = I_n,$$

which shows that A is invertible and that

$$A^{-1} = A^T.$$

Since A has real entries, we have $A^* = A^T$, and hence,

$$A^*A = A^T A = I_n.$$

Similarly, we have

$$AA^* = AA^T = I_n.$$

This shows that any orthogonal matrix is a unitary matrix.

The converse is not true in general. For instance, the matrix

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

is a unitary matrix but not an orthogonal matrix. $\square$

Exercise 3.59. Determine whether the following matrices are unitary, hermitian, both, or neither.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Exercise 3.60. Let z, w be complex numbers such that $|z|^2 + |w|^2 = 1$. Show that

$$\begin{pmatrix} z & -\overline{w} \\ w & \overline{z} \end{pmatrix}$$

is a unitary matrix.

Exercise 3.61. Let A be a diagonal matrix with complex entries. Show that A is a unitary matrix if and only if all the diagonal entries of A have absolute value equal to 1.



Definition

A square matrix A with complex entries is called **normal** if

$$AA^* = A^*A,$$

where A^* denotes the conjugate transpose of A .



Example

The following matrices are normal.

$$\begin{pmatrix} 1 & 2+i \\ 2-i & 3 \end{pmatrix}, \begin{pmatrix} 5 & i \\ -i & 7 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Some non-normal matrices are as follows.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 3.62. Show that any unitary matrix is a normal matrix. Is the converse true?

Exercise 3.63. Show that any hermitian matrix is a normal matrix. Is the converse true?

Exercise 3.64. Show that any skew-hermitian matrix is a normal matrix. Is the converse true?

Here is a summary of some properties of some of the special types of matrices introduced above.

Matrix	Property
Diagonal	Entries outside the main diagonal are zero
Scalar	Diagonal matrix with all diagonal entries equal
Triangular	Entries below or above the main diagonal are zero
Upper-triangular	Entries below the main diagonal are zero
Lower-triangular	Entries above the main diagonal are zero
Symmetric	$A^T = A$
Skew-symmetric	$A^T = -A$
Orthogonal	$AA^T = A^T A = I_n$
Hermitian	$A^* = A$
Skew-hermitian	$A^* = -A$
Unitary	$AA^* = A^* A = I_n$
Normal	$AA^* = A^* A$

